

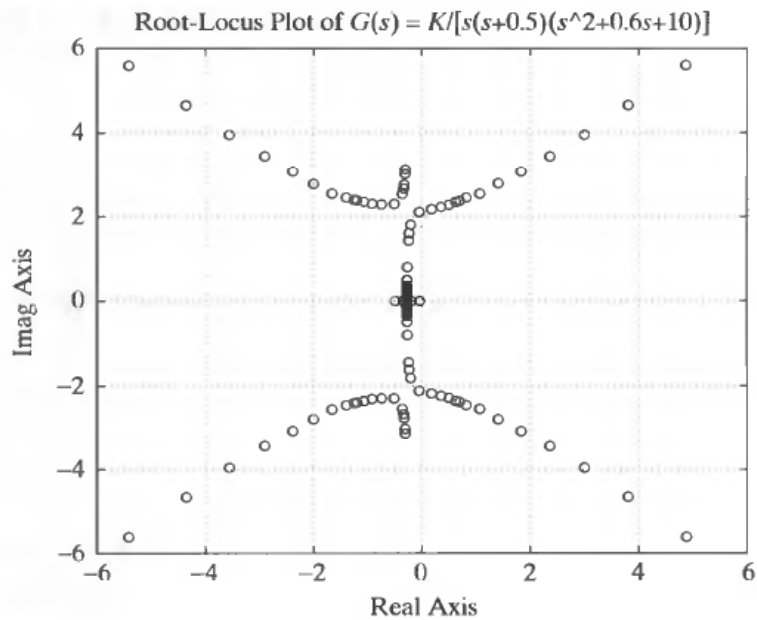


Figure 6–21
Root-locus plot.

MATLAB Program 6–6

```
% ----- Root-locus plot -----

num = [0 0 0 0 1];
den = [1 1.1 10.3 5 0];
K1 = 0:0.2:20;
K2 = 20:0.1:30;
K3 = 30:5:1000;
K = [K1 K2 K3];
r = rlocus(num,den,K);
plot(r,'ob')
v = [-4 4 -4 4]; axis(v)
grid
title('Root-Locus Plot of  $G(s) = K/[s(s + 0.5)(s^2 + 0.6s + 10)]$ ')
xlabel('Real Axis')
ylabel('Imag Axis')

% ***** Note that the command 'plot(r,'ob')' gives small circles
% in the screen plot in blue color *****
```

EXAMPLE 6–7 Consider the system shown in Figure 6–23. The system equations are

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}u$$

$$y = \mathbf{C}\mathbf{x} + Du$$

$$u = r - y$$

In this example problem we shall obtain the root-locus diagram of the system defined in state space. Let us assume, for example, that matrices **A**, **B**, **C**, and **D** are given by

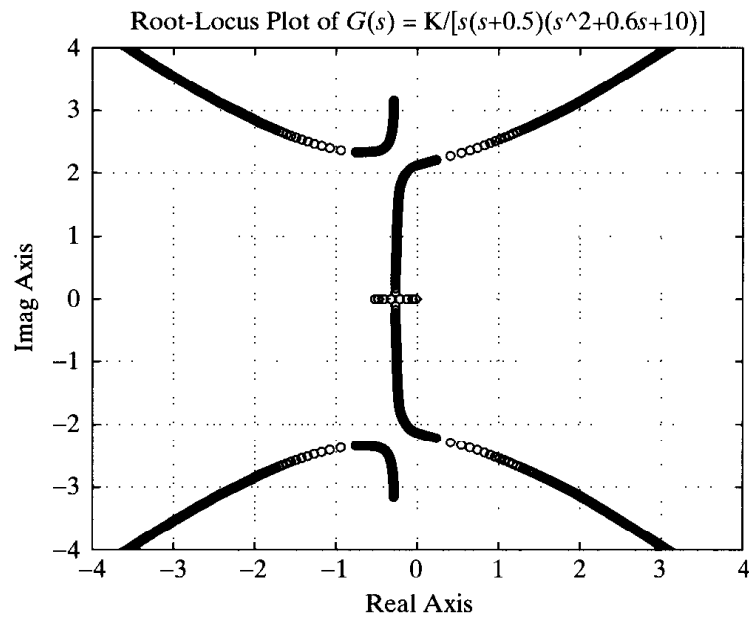


Figure 6-22
Root-locus plot.

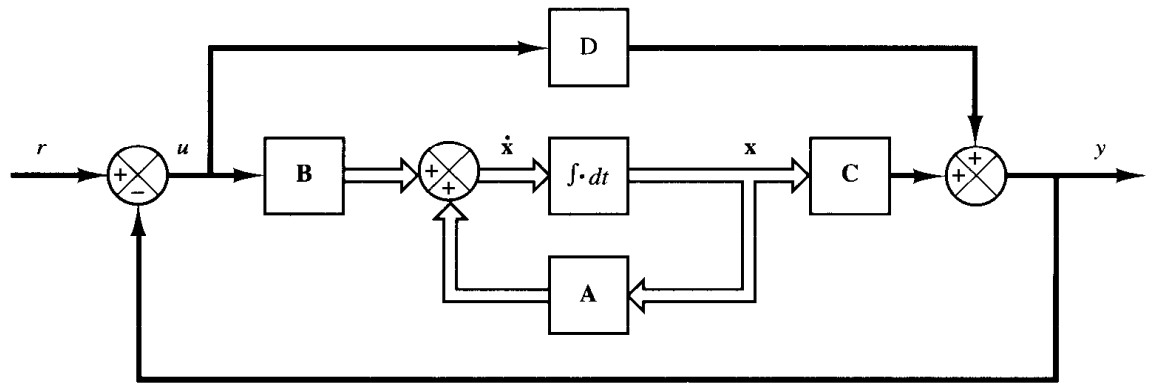


Figure 6-23
Closed-loop control
system.

$$\mathbf{A} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -160 & -56 & -14 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 0 \\ 1 \\ -14 \end{bmatrix} \quad (6-18)$$

$$\mathbf{C} = [1 \quad 0 \quad 0], \quad \mathbf{D} = [0]$$

The root-locus plot for this system can be obtained with MATLAB by use of the following command:

rlocus(A,B,C,D)

This command will produce the same root-locus plot as can be obtained by use of the rlocus (num,den) command, where num and den are obtained from

[num,den] = ss2tf(A,B,C,D)

as follows:

num = [0 0 1 0]
den = [1 14 56 160]

MATLAB Program 6–7 gives a program that will generate the root-locus plot as shown in Figure 6–24.

```

MATLAB Program 6–7

% ----- Root-locus plot -----

% ***** Root-locus plot of system defined in state space *****

% ***** Enter matrices A, B, C, and D *****

A = [0 1 0; 0 0 1; -160 -56 -14];
B = [0; 1; -14];
C = [1 0 0];
D = [0];

% ***** Enter rlocus(A,B,C,D) command in the computer *****

rlocus(A,B,C,D)
grid
title('Root-Locus Plot of System defined in State Space')

```

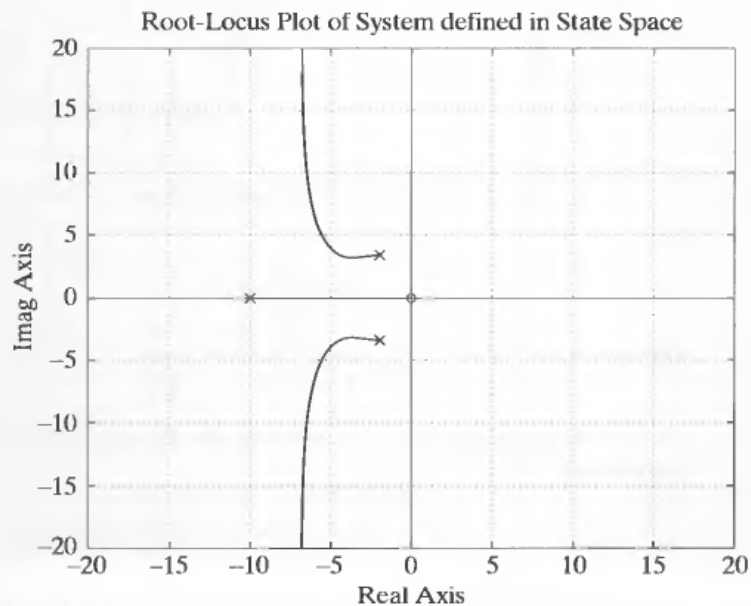


Figure 6–24

Root-locus plot of system defined in state space, where **A**, **B**, **C**, and **D** are as given by Equation (6–18).

6–5 SPECIAL CASES

In this section we shall consider two special cases. One is the case in which the gain K does not appear as a multiplicative factor, and in the other the closed-loop system is a positive-feedback system, rather than a negative-feedback system.

Constructing root loci when a variable parameter does not appear as a multiplying factor. In some cases the variable parameter K may not appear as a multiplying factor of $G(s)H(s)$. In such cases it may be possible to rewrite the characteristic equation such that the variable parameter K appears as a multiplying factor of $G(s)H(s)$. Example 6–8 illustrates how to proceed in such a case.

EXAMPLE 6–8

Consider the system in Figure 6–25. Draw a root-locus diagram. Then determine the value of k such that the damping ratio of the dominant closed-loop poles is 0.4.

Here the system involves velocity feedback. The open-loop transfer function is

$$\text{Open-loop transfer function} = \frac{20}{s(s+1)(s+4) + 20ks}$$

Notice that the adjustable variable k does not appear as a multiplying factor. The characteristic equation for the system is

$$s^3 + 5s^2 + 4s + 20 + 20ks = 0 \quad (6-19)$$

Define

$$20k = K$$

Then Equation (6–19) becomes

$$s^3 + 5s^2 + 4s + Ks + 20 = 0 \quad (6-20)$$

Dividing both sides of Equation (6–20) by the sum of the terms that do not contain K , we get

$$1 + \frac{Ks}{s^3 + 5s^2 + 4s + 20} = 0$$

or

$$1 + \frac{Ks}{(s+j2)(s-j2)(s+5)} = 0 \quad (6-21)$$

Equation (6–21) is now of the form of Equation (6–5).

We shall now sketch the root loci of the system given by Equation (6–21). Notice that the open-loop poles are located at $s = j2, s = -j2, s = -5$, and the open-loop zero is located at $s = 0$. The root locus exists on the real axis between 0 and -5 . Since

$$\lim_{s \rightarrow \infty} \frac{Ks}{(s+j2)(s-j2)(s+5)} = \lim_{s \rightarrow \infty} \frac{K}{s^2}$$

we have

$$\text{Angle of asymptote} = \frac{\pm 180^\circ(2k+1)}{2} = \pm 90^\circ$$

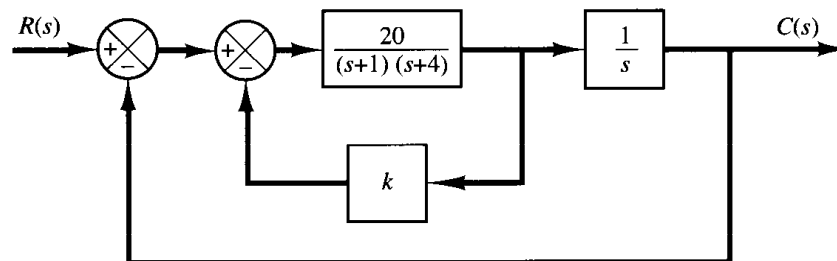


Figure 6–25
Control system.

The intersection of the asymptotes with the real axis can be found from

$$\lim_{s \rightarrow \infty} \frac{Ks}{s^3 + 5s^2 + 4s + 20} = \lim_{s \rightarrow \infty} \frac{K}{s^2 + 5s + \dots} = \lim_{s \rightarrow \infty} \frac{K}{(s + 2.5)^2}$$

as

$$\sigma_a = -2.5$$

The angle of departure (angle θ) from the pole at $s = j2$ is obtained as follows:

$$\theta = 180^\circ - 90^\circ - 21.8^\circ + 90^\circ = 158.2^\circ$$

Thus, the angle of departure from the pole $s = j2$ is 158.2° . Figure 6–26 shows a root-locus plot for the system.

Note that the closed-loop poles with $\zeta = 0.4$ must lie on straight lines passing through the origin and making the angles $\pm 66.42^\circ$ with the negative real axis. In the present case, there are two intersections of the root-locus branch in the upper half s plane and the straight line of angle 66.42° . Thus, two values of K will give the damping ratio ζ of the closed-loop poles equal to 0.4. At point P , the value of K is

$$K = \left| \frac{(s + j2)(s - j2)(s + 5)}{s} \right|_{s=-1.0490+j2.4065} = 8.9801$$

Hence

$$k = \frac{K}{20} = 0.4490 \quad \text{at point } P$$

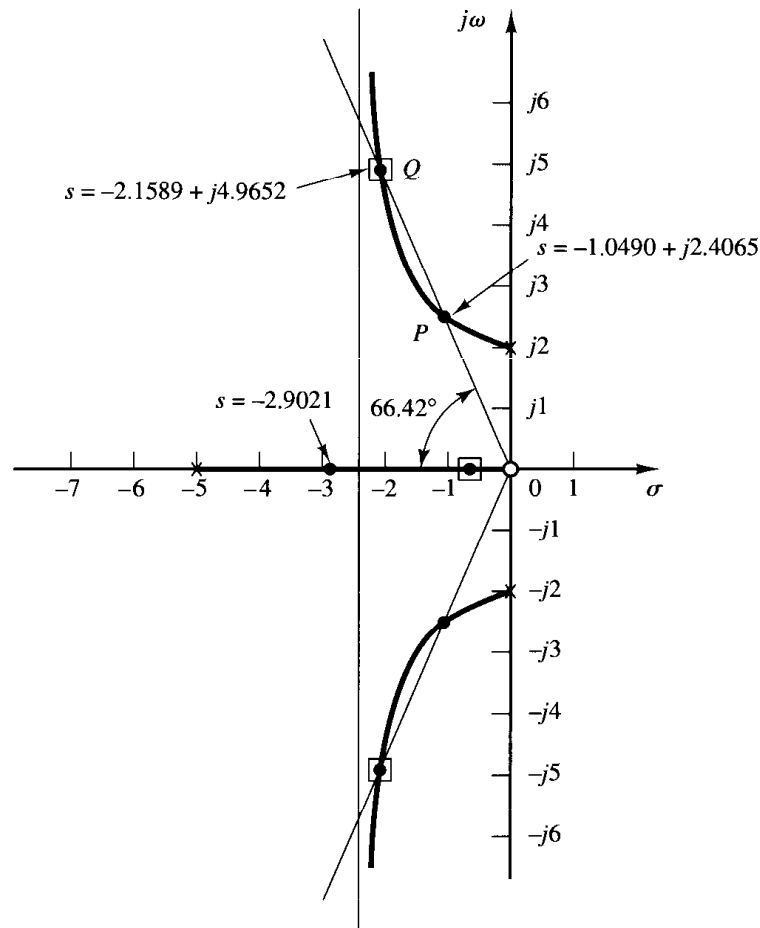


Figure 6–26
Root-locus plot for
the system shown in
Figure 6–25.

At point Q , the value of K is

$$K = \left| \frac{(s + j2)(s - j2)(s + 5)}{s} \right|_{s = -2.1589 + j4.9652} = 28.260$$

Hence

$$k = \frac{K}{20} = 1.4130 \quad \text{at point } Q$$

Thus, we have two solutions for this problem. For $k = 0.4490$, the three closed-loop poles are located at

$$s = -1.0490 + j2.4065, \quad s = -1.0490 - j2.4065, \quad s = -2.9021$$

For $k = 1.4130$, the three closed-loop poles are located at

$$s = -2.1589 + j4.9652, \quad s = -2.1589 - j4.9652, \quad s = -0.6823$$

It is important to point out that the zero at the origin is the open-loop zero, but not the closed-loop zero. This is evident, because the original system shown in Figure 6-25 does not have a closed-loop zero, since

$$\frac{C(s)}{R(s)} = \frac{20}{s(s + 1)(s + 4) + 20(1 + ks)}$$

The open-loop zero at $s = 0$ was introduced in the process of modifying the characteristic equation such that the adjustable variable $K = 20k$ was to appear as a multiplying factor.

We have obtained two different values of k to satisfy the requirement that the damping ratio of the dominant closed-loop poles be equal to 0.4. The closed-loop transfer function with $k = 0.4490$ is given by

$$\begin{aligned} \frac{C(s)}{R(s)} &= \frac{20}{s^3 + 5s^2 + 12.98s + 20} \\ &= \frac{20}{(s + 1.0490 + j2.4065)(s + 1.0490 - j2.4065)(s + 2.9021)} \end{aligned}$$

The closed-loop transfer function with $k = 1.4130$ is given by

$$\begin{aligned} \frac{C(s)}{R(s)} &= \frac{20}{s^3 + 5s^2 + 32.26s + 20} \\ &= \frac{20}{(s + 2.1589 + j4.9652)(s + 2.1589 - j4.9652)(s + 0.6823)} \end{aligned}$$

Notice that the system with $k = 0.4490$ has a pair of dominant complex-conjugate closed-loop poles, while in the system with $k = 1.4130$ the real closed-loop pole at $s = -0.6823$ is dominant, and the complex-conjugate closed-loop poles are not dominant. In this case, the response characteristic is primarily determined by the real closed-loop pole.

Let us compare the unit-step responses of both systems. MATLAB Program 6-8 may be used for plotting the unit-step response curves in one diagram. The resulting unit-step response curves $[c_1(t) \text{ for } k = 0.4490 \text{ and } c_2(t) \text{ for } k = 1.4130]$ are shown in Figure 6-27.

From Figure 6-27 we notice that the response of the system with $k = 0.4490$ is oscillatory. (The effect of the closed-loop pole at $s = -2.9021$ on the unit-step response is small.) For the system with $k = 1.4130$, the oscillations due to the closed-loop poles at $s = -2.1589 \pm j4.9652$ damp out much faster than purely exponential response due to the closed-loop pole at $s = -0.6823$.

MATLAB Program 6-8

```
% ----- Unit-step response -----

% ***** Enter numerators and denominators of system with
% k = 0.4490 and k = 1.4130, respectively. *****

num1 = [0 0 0 20];
den1 = [1 5 12.98 20];
num2 = [0 0 0 20];
den2 = [1 5 32.26 20];
t = 0:0.1:10;
[c1,x1,t] = step(num1,den1,t);
[c2,x2,t] = step(num2,den2,t);
plot(t,c1,t,c2)
text(2.5,1.12,'k = 0.4490')
text(3.7,0.85,'k = 1.4130')
grid
title('Unit-Step Responses of Two Systems')
xlabel('t Sec')
ylabel('Outputs c1 and c2')
```

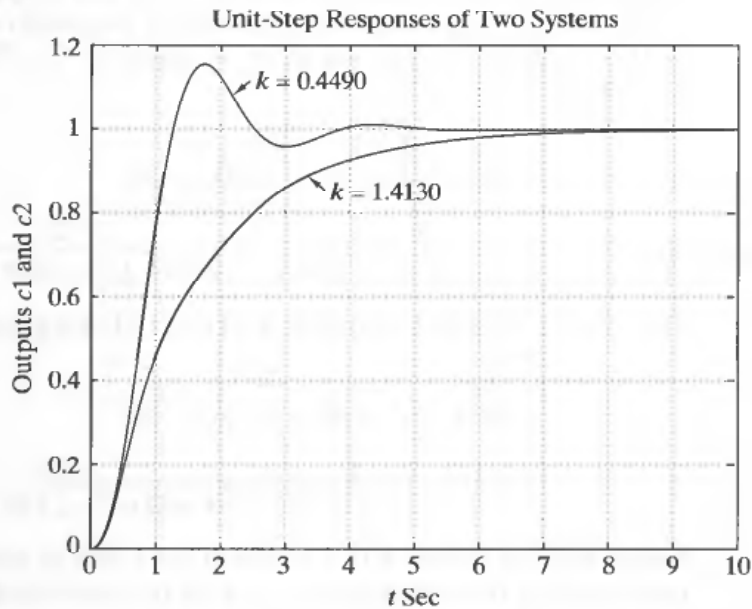


Figure 6-27
Unit-step response curves for the system shown in Figure 6-25 when the damping ratio ζ of the dominant closed-loop poles is set equal to 0.4. (Two possible values of k give the damping ratio ζ equal to 0.4.)

The system with $k = 0.4490$ (which exhibits a faster response with relatively small overshoot) has a much better response characteristic than the system with $k = 1.4130$ (which exhibits a slow overdamped response). Therefore, we should choose $k = 0.4490$ for the present system.

Root loci for positive-feedback systems.* In a complex control system, there may be a positive-feedback inner loop as shown in Figure 6-28. Such a loop is usu-

* Reference W-5.

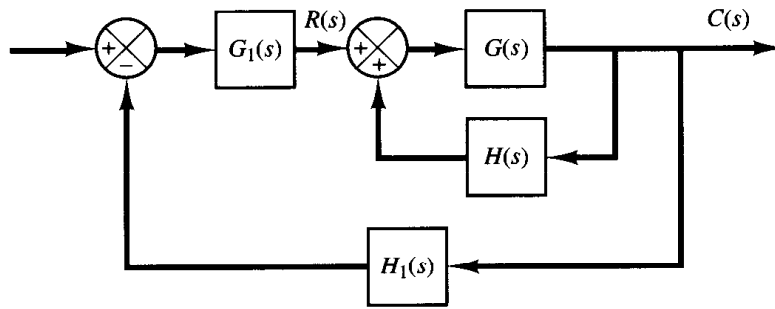


Figure 6–28
Control system.

ally stabilized by the outer loop. In what follows, we shall be concerned only with the positive-feedback inner loop. The closed-loop transfer function of the inner loop is

$$\frac{C(s)}{R(s)} = \frac{G(s)}{1 - G(s)H(s)}$$

The characteristic equation is

$$1 - G(s)H(s) = 0 \quad (6-22)$$

This equation can be solved in a manner similar to the development of the root-locus method in Section 6–2. The angle condition, however, must be altered.

Equation (6–22) can be rewritten as

$$G(s)H(s) = 1$$

which is equivalent to the following two equations:

$$\angle G(s)H(s) = 0^\circ \pm k360^\circ \quad (k = 0, 1, 2, \dots)$$

$$|G(s)H(s)| = 1$$

The total sum of all angles from the open-loop poles and zeros must be equal to $0^\circ \pm k360^\circ$. Thus the root locus follows a 0° locus in contrast to the 180° locus considered previously. The magnitude condition remains unaltered.

To illustrate the root-locus plot for the positive feedback system, we shall use the following transfer functions $G(s)$ and $H(s)$ as an example.

$$G(s) = \frac{K(s + 2)}{(s + 3)(s^2 + 2s + 2)}, \quad H(s) = 1$$

The gain K is assumed to be positive.

The general rules for constructing root loci given in Section 6–3 must be modified in the following way:

Rule 2 is modified as follows: If the total number of real poles and real zeros to the right of a test point on the real axis is even, then this test point lies on the root locus.

Rule 3 is modified as follows:

$$\text{Angles of asymptotes} = \frac{\pm k 360^\circ}{n - m} \quad (k = 0, 1, 2, \dots)$$

where n = number of finite poles of $G(s)H(s)$

m = number of finite zeros of $G(s)H(s)$

Rule 5 is modified as follows: When calculating the angle of departure (or angle of arrival) from a complex open-loop pole (or at a complex zero), subtract from 0° the sum of all angles of the vectors from all the other poles and zeros to the complex pole (or complex zero) in question, with appropriate signs included.

Other rules for constructing the root-locus plot remain the same. We shall now apply the modified rules to construct the root-locus plot.

1. Plot the open-loop poles ($s = -1 + j$, $s = -1 - j$, $s = -3$) and zero ($s = -2$) in the complex plane. As K is increased from 0 to ∞ , the closed-loop poles start at the open-loop poles and terminate at the open-loop zeros (finite or infinite), just as in the case of negative-feedback systems.
2. Determine the root loci on the real axis. Root loci exist on the real axis between -2 and $+\infty$ and between -3 and $-\infty$.
3. Determine the asymptotes of the root loci. For the present system,

$$\text{Angle of asymptote} = \frac{\pm k 360^\circ}{3 - 1} = \pm 180^\circ$$

This simply means that asymptotes are on the real axis.

4. Determine the breakaway and break-in points. Since the characteristic equation is

$$(s + 3)(s^2 + 2s + 2) - K(s + 2) = 0$$

we obtain

$$K = \frac{(s + 3)(s^2 + 2s + 2)}{s + 2}$$

By differentiating K with respect to s , we obtain

$$\frac{dK}{ds} = \frac{2s^3 + 11s^2 + 20s + 10}{(s + 2)^2}$$

Note that

$$\begin{aligned} 2s^3 + 11s^2 + 20s + 10 &= 2(s + 0.8)(s^2 + 4.7s + 6.24) \\ &= 2(s + 0.8)(s + 2.35 + j0.77)(s + 2.35 - j0.77) \end{aligned}$$

Point $s = -0.8$ is on the root locus. Since this point lies between two zeros (a finite zero and an infinite zero), it is an actual break-in point. Points $s = -2.35 \pm j0.77$ do not satisfy the angle condition and, therefore, they are neither breakaway nor break-in points.

5. Find the angle of departure of the root locus from a complex pole. For the complex pole at $s = -1 + j$, the angle of departure θ is

$$\theta = 0^\circ - 27^\circ - 90^\circ + 45^\circ$$

or

$$\theta = -72^\circ$$

(The angle of departure from the complex pole at $s = -1 - j$ is 72° .)

6. Choose a test point in the broad neighborhood of the $j\omega$ axis and the origin and apply the angle condition. Locate a sufficient number of points that satisfy the angle condition.

Figure 6-29 shows the root loci for the given positive-feedback system. The root loci are shown with dashed lines and curve.

Note that if

$$K > \left. \frac{(s+3)(s^2+2s+2)}{s+2} \right|_{s=0} = 3$$

one real root enters the right-half s plane. Hence, for values of K greater than 3, the system becomes unstable. (For $K > 3$, the system must be stabilized with an outer loop.)

Note that the closed-loop transfer function for the positive-feedback system is given by

$$\begin{aligned} \frac{C(s)}{R(s)} &= \frac{G(s)}{1 - G(s)H(s)} \\ &= \frac{K(s+2)}{(s+3)(s^2+2s+2) - K(s+2)} \end{aligned}$$

To compare this root-locus plot with that of the corresponding negative-feedback system, we show in Figure 6-30 the root loci for the negative-feedback system whose closed-loop transfer function is

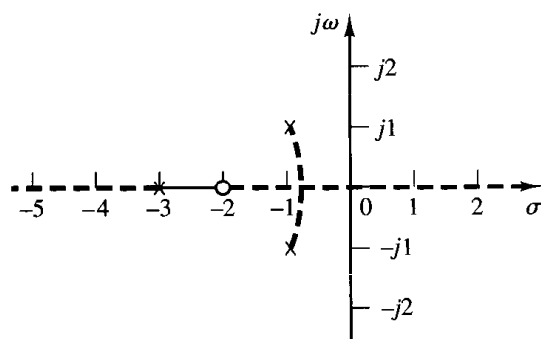


Figure 6-29
Root-locus plot for the positive-feedback system with $G(s) = K(s+2)/[(s+3)(s^2+2s+2)]$, $H(s) = 1$.

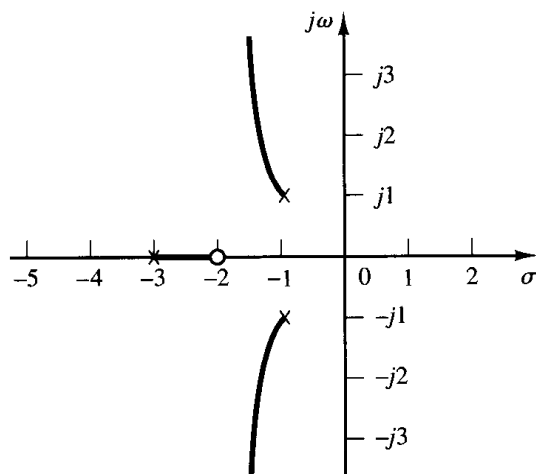
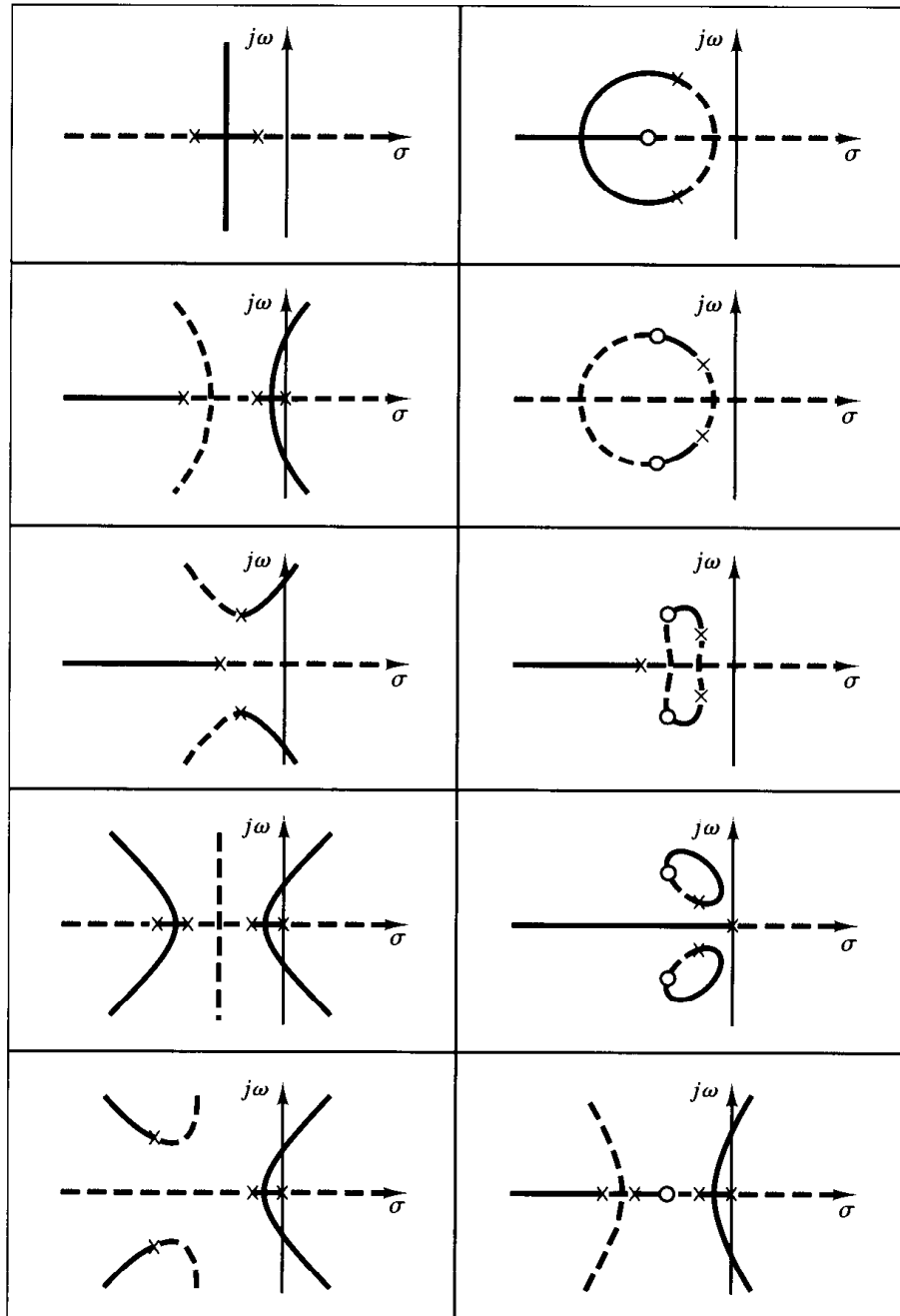


Figure 6-30
Root-locus plot for the negative-feedback system with $G(s) = K(s+2)/[(s+3)(s^2+2s+2)]$, $H(s) = 1$.

$$\frac{C(s)}{R(s)} = \frac{K(s + 2)}{(s + 3)(s^2 + 2s + 2) + K(s + 2)}$$

Table 6-2 shows various root-locus plots of negative-feedback and positive-feedback systems. The closed-loop transfer functions are given by

Table 6-2 Root-Locus Plots of Negative-Feedback and Positive-Feedback Systems



Heavy lines and curves correspond to negative-feedback systems; dashed lines and curves correspond to positive-feedback systems.

$$\frac{C}{R} = \frac{G}{1 + GH}, \quad \text{for negative-feedback systems}$$

$$\frac{C}{R} = \frac{G}{1 - GH}, \quad \text{for positive-feedback systems}$$

where GH is the open-loop transfer function. In Table 6-2, the root loci for negative-feedback systems are drawn with heavy lines and curves and those for positive-feedback systems are drawn with dashed lines and curves.

6-6 ROOT-LOCUS ANALYSIS OF CONTROL SYSTEMS

In this section we shall first discuss orthogonality of the root loci and constant-gain loci for the closed-loop systems. Next, we discuss conditionally stable systems. Finally, we analyze nonminimum-phase systems.

Orthogonality of root loci and constant-gain loci. Consider the system whose open-loop transfer function is $G(s)H(s)$. In the $G(s)H(s)$ plane, the loci of $|G(s)H(s)| = \text{constant}$ are circles centered at the origin, and the loci corresponding to $\angle G(s)H(s) = \pm 180^\circ(2k + 1)$ ($k = 0, 1, 2, \dots$) lie on the negative real axis of the $G(s)H(s)$ plane, as shown in Figure 6-31. [Note that the complex plane employed here is not the s plane, but the $G(s)H(s)$ plane.]

The root loci and constant-gain loci in the s plane are conformal mappings of the loci of $\angle G(s)H(s) = \pm 180^\circ(2k + 1)$ and of $|G(s)H(s)| = \text{constant}$ in the $G(s)H(s)$ plane.

Since the constant-phase and constant-gain loci in the $G(s)H(s)$ plane are orthogonal, the root loci and constant-gain loci in the s plane are orthogonal. Figure 6-32(a) shows the root loci and constant-gain loci for the following system:

$$G(s) = \frac{K(s + 2)}{s^2 + 2s + 3}, \quad H(s) = 1$$

Notice that since the pole-zero configuration is symmetrical about the real axis the constant-gain loci are also symmetrical about the real axis.

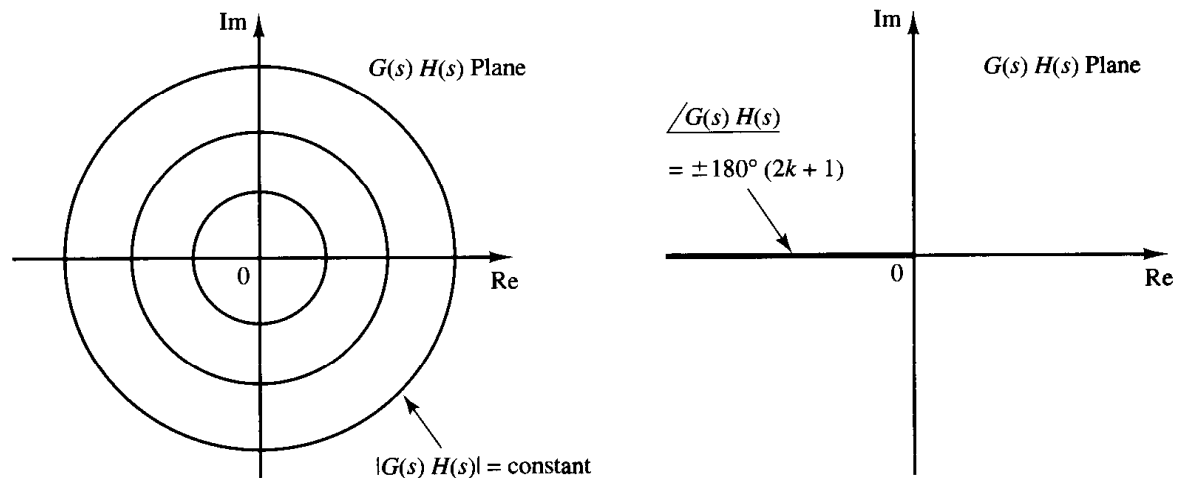


Figure 6-31
Plots of constant-gain and constant-phase loci in the $G(s)H(s)$ plane.

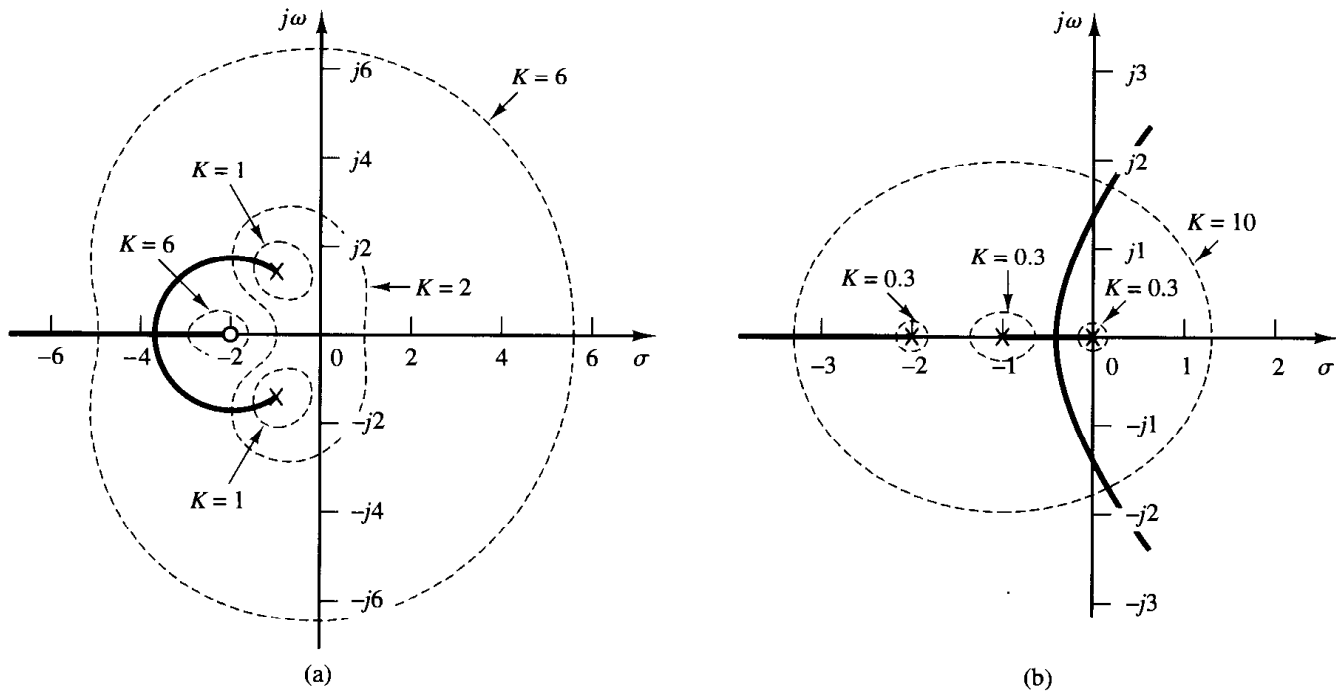


Figure 6-32

Plots of root loci and constant-gain loci. (a) System with $G(s) = K(s + 2)/(s^2 + 2s + 3)$, $H(s) = 1$; (b) system with $G(s) = K/[s(s + 1)(s + 2)]$, $H(s) = 1$.

Figure 6-32(b) shows the root loci and constant-gain loci for the system:

$$G(s) = \frac{K}{s(s + 1)(s + 2)}, \quad H(s) = 1$$

Notice that since the configuration of the poles in the s plane is symmetrical about the real axis and the line parallel to the imaginary axis passing through point $(\sigma = -1, \omega = 0)$, the constant-gain loci are symmetrical about the $\omega = 0$ line (real axis) and the $\sigma = -1$ line.

Conditionally stable systems. Consider the system shown in Figure 6-33(a). The root loci for this system can be plotted by applying the general rules and procedure for constructing root loci. A root-locus plot for this system is shown in Figure 6-33(b). It can be seen that this system is stable only for limited ranges of the value of K ; that is, $0 < K < 14$ and $64 < K < 195$. The system becomes unstable for $14 < K < 64$ and $195 < K$. If K assumes a value corresponding to unstable operation, the system may break down or may become nonlinear due to a saturation nonlinearity that may exist. Such a system is called *conditionally stable*.

In practice, conditionally stable systems are not desirable. Conditional stability is dangerous but does occur in certain systems, in particular, a system that has an unstable feedforward path. Such a feedforward path may occur if the system has a minor loop. It is advisable to avoid such conditional stability since, if the gain drops beyond the critical value for some reason, the system becomes unstable. Note that the addition of a proper compensating network will eliminate conditional stability. [An addition of a zero will cause the root loci to bend to the left. (See Section 7-2). Hence conditional stability may be eliminated by adding proper compensation.]

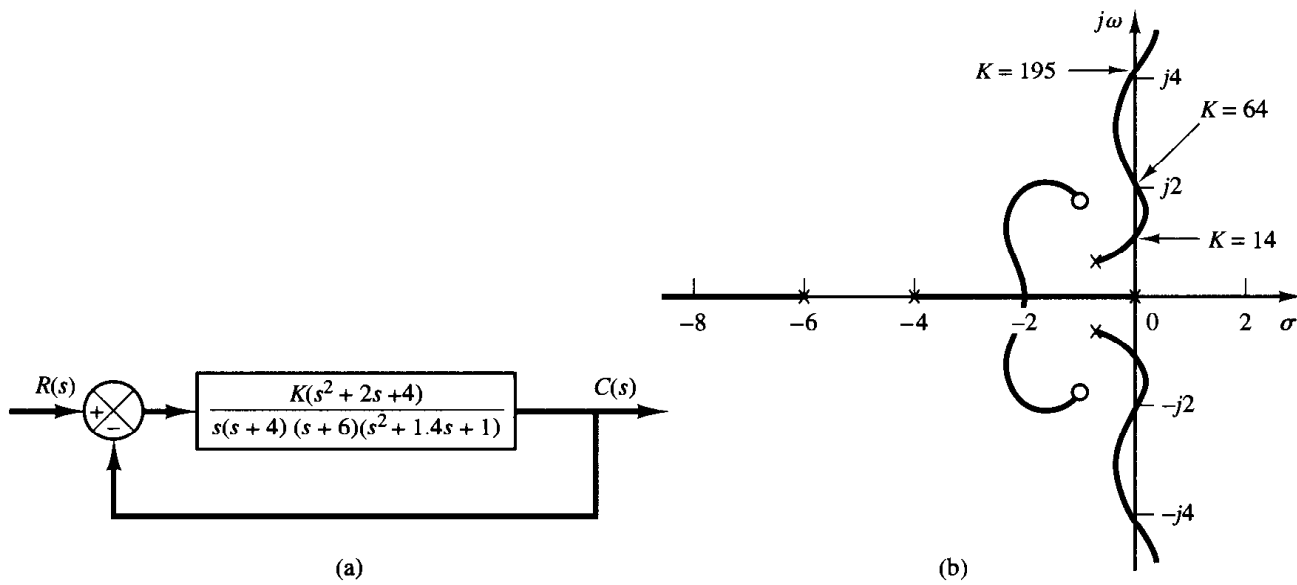


Figure 6-33

(a) Conditionally stable system; (b) root-locus plot.

Nonminimum-phase systems. If all the poles and zeros of a system lie in the left-half s plane, then the system is called *minimum phase*. If a system has at least one pole or zero in the right-half s plane, then the system is called *nonminimum phase*. The term nonminimum phase comes from the phase shift characteristics of such a system when subjected to sinusoidal inputs.

Consider the system shown in Figure 6-34(a). For this system

$$G(s) = \frac{K(1 - T_a s)}{s(Ts + 1)} \quad (T_a > 0), \quad H(s) = 1$$

This is a nonminimum-phase system since there is one zero in the right-half s plane. For this system, the angle condition becomes

$$\angle G(s) = \angle -\frac{K(T_a s - 1)}{s(Ts + 1)}$$

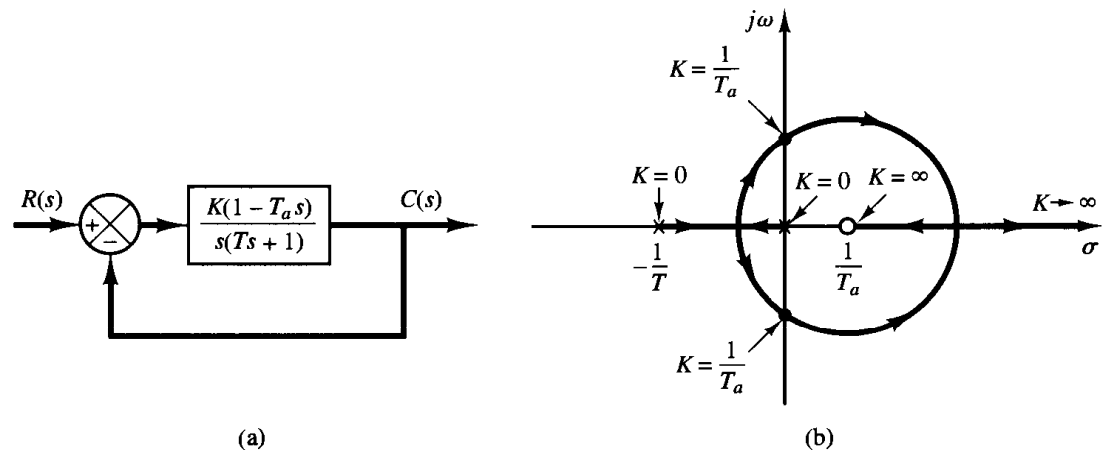


Figure 6-34

(a) Nonminimum phase system; (b) root-locus plot.

$$\begin{aligned}
 &= \angle \frac{K(T_a s - 1)}{s(Ts + 1)} + 180^\circ \\
 &= \pm 180^\circ(2k + 1) \quad (k = 0, 1, 2, \dots)
 \end{aligned}$$

or

$$\angle \frac{K(T_a s - 1)}{s(Ts + 1)} = 0^\circ \quad (6-23)$$

The root loci can be obtained from Equation (6-23). Figure 6-34(b) shows a root-locus plot for this system. From the diagram, we see that the system is stable if the gain K is less than $1/T_a$.

6-7 ROOT LOCI FOR SYSTEMS WITH TRANSPORT LAG

Figure 6-35 shows a thermal system in which hot air is circulated to keep the temperature of a chamber constant. In this system, the measuring element is placed downstream a distance L ft from the furnace, the air velocity is v ft/sec, and $T = L/v$ sec would elapse before any change in the furnace temperature is sensed by the thermometer. Such a delay in measuring, delay in controller action, or delay in actuator operation, and the like, is called *transport lag* or *dead time*. Dead time is present in most process control systems.

The input $x(t)$ and the output $y(t)$ of a transport lag or dead time element are related by

$$y(t) = x(t - T)$$

where T is dead time. The transfer function of transport lag or dead time is given by

$$\begin{aligned}
 \text{Transfer function of transport lag or dead time} &= \frac{\mathcal{L}[x(t - T)1(t - T)]}{\mathcal{L}[x(t)1(t)]} \\
 &= \frac{X(s)e^{-Ts}}{X(s)} = e^{-Ts}
 \end{aligned}$$

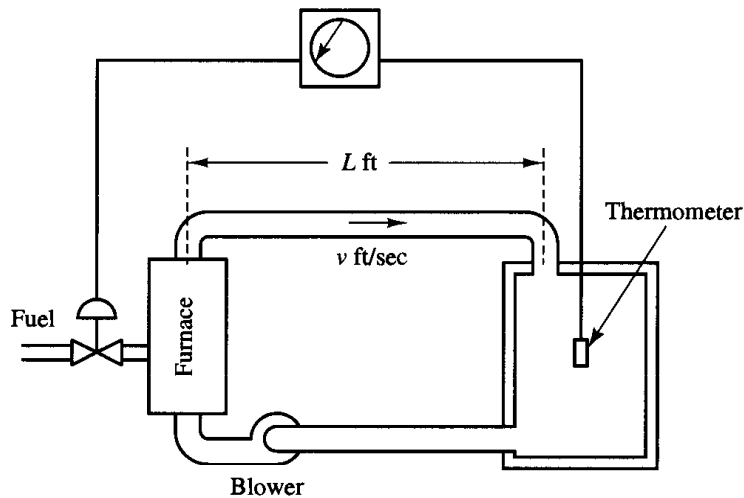


Figure 6-35
Thermal system.

Suppose that the feedforward transfer function of this thermal system can be approximated by

$$G(s) = \frac{Ke^{-Ts}}{s + 1}$$

as shown in Figure 6–36. Let us construct a root-locus plot for this system. The characteristic equation for this closed-loop system is

$$1 + \frac{Ke^{-Ts}}{s + 1} = 0 \quad (6-24)$$

It is noted that for systems with transport lag the rules of construction presented earlier need to be modified. For example, the number of the root-locus branches is infinite, since the characteristic equation has an infinite number of roots. The number of asymptotes is infinite. They are all parallel to the real axis of the s plane.

From Equation (6–24), we obtain

$$\frac{Ke^{-Ts}}{s + 1} = -1$$

Thus, the angle condition becomes

$$\angle \frac{Ke^{-Ts}}{s + 1} = \angle e^{-Ts} - \angle s + 1 = \pm 180^\circ(2k + 1) \quad (k = 0, 1, 2, \dots) \quad (6-25)$$

To find the angle of e^{-Ts} , substitute $s = \sigma + j\omega$. Then we obtain

$$e^{-Ts} = e^{-T\sigma - j\omega T}$$

Since $e^{-T\sigma}$ is a real quantity, the angle of $e^{-T\sigma}$ is zero. Hence

$$\begin{aligned} \angle e^{-Ts} &= \angle e^{-j\omega T} = \angle \cos \omega T - j \sin \omega T \\ &= -\omega T \quad (\text{radians}) \\ &= -57.3\omega T \quad (\text{degrees}) \end{aligned}$$

The angle condition, Equation (6–25), then becomes

$$-57.3\omega T - \angle s + 1 = \pm 180^\circ(2k + 1)$$

Since T is a given constant, the angle of e^{-Ts} is a function of ω only.

We shall next determine the angle contribution due to e^{-Ts} . For $k = 0$, the angle condition may be written

$$\angle s + 1 = \pm 180^\circ - 57.3^\circ \omega T \quad (6-26)$$

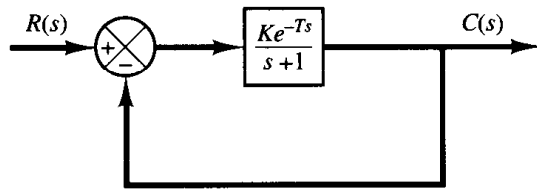


Figure 6–36
Block diagram of the system shown in Figure 6–35.

Since the angle contribution of e^{-Ts} is zero for $\omega = 0$, the real axis from -1 to $-\infty$ forms a part of the root loci. Now assume a value ω_1 for ω and compute $57.3^\circ\omega_1 T$. At point -1 on the negative real axis, draw a line that makes an angle of $180^\circ - 57.3^\circ\omega_1 T$ with the real axis. Find the intersection of this line and the horizontal line $\omega = \omega_1$. This intersection, point P in Figure 6-37(a), is a point satisfying Equation (6-26) and hence is on a root locus. Continuing the same process, we obtain the root-locus plot as shown in Figure 6-37(b).

Note that as s approaches minus infinity, the open-loop transfer function

$$\frac{Ke^{-Ts}}{s+1}$$

approaches minus infinity since

$$\begin{aligned}\lim_{s=-\infty} \frac{Ke^{-Ts}}{s+1} &= \frac{\frac{d}{ds}(Ke^{-Ts})}{\frac{d}{ds}(s+1)} \bigg|_{s=-\infty} \\ &= -KTe^{-Ts} \bigg|_{s=-\infty} \\ &= -\infty\end{aligned}$$

Therefore, $s = -\infty$ is a pole of the open-loop transfer function. Thus, root loci start from $s = -1$ or $s = -\infty$ and terminate at $s = \infty$, as K increases from zero to infinity. Since the right-hand side of the angle condition given by Equation (6-25) has an infinite number of values, there are an infinite number of root loci, as the value of k ($k = 0, 1, 2, \dots$) goes from zero to infinity. For example, if $k = 1$, the angle condition becomes

$$\begin{aligned}\angle s+1 &= \pm 540^\circ - 57.3^\circ\omega T \quad (\text{degrees}) \\ &= \pm 3\pi - \omega T \quad (\text{radians})\end{aligned}$$

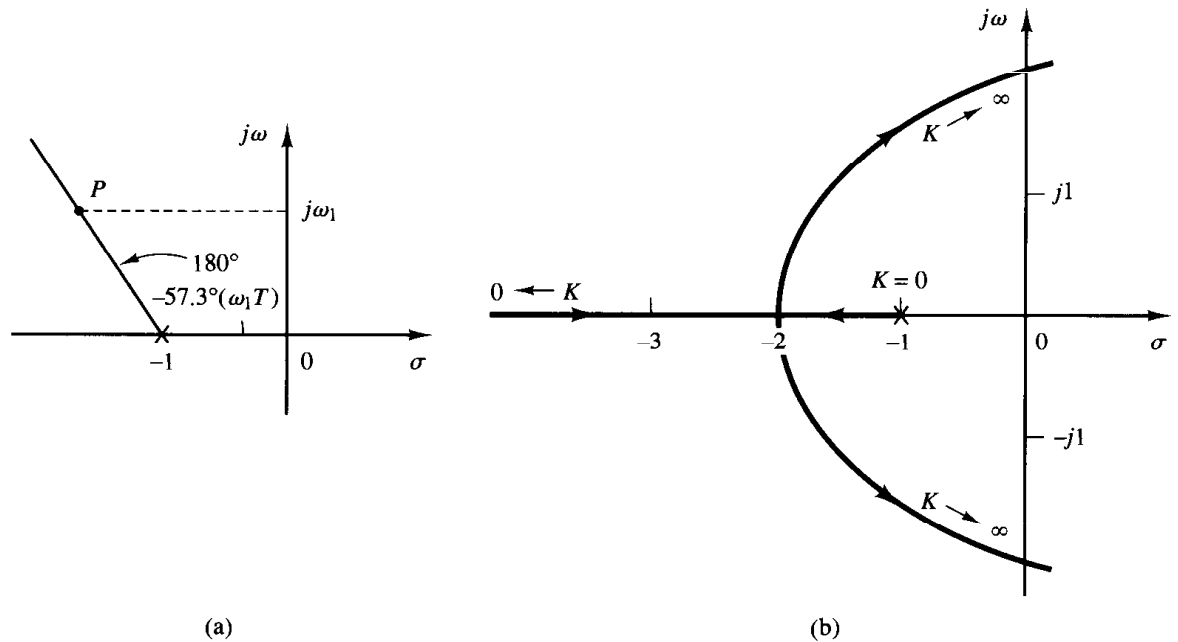


Figure 6-37
(a) Construction of the root locus; (b) root-locus plot.

The construction of the root loci for $k = 1$ is the same as that for $k = 0$. A plot of root loci for $k = 0, 1$, and 2 when $T = 1$ sec is shown in Figure 6–38.

The magnitude condition states that

$$\left| \frac{Ke^{-Ts}}{s + 1} \right| = 1$$

Since the magnitude of e^{-Ts} is equal to that of $e^{-T\sigma}$ or

$$|e^{-Ts}| = |e^{-T\sigma}| \cdot |e^{-j\omega T}| = e^{-T\sigma}$$

the magnitude condition becomes

$$|s + 1| = Ke^{-T\sigma}$$

The root loci shown in Figure 6–38 are graduated in terms of K when $T = 1$ sec.

Although there are an infinite number of root-locus branches, the primary branch that lies between $-j\pi$ and $j\pi$ is most important. Referring to Figure 6–38, the critical value of K at the primary branch is equal to 2, while the critical values of K at other branches are much higher (8, 14, . . .). Therefore, the critical value $K = 2$ on the primary branch is most significant from the stability viewpoint. The transient response of the system is determined by the roots located closest to the $j\omega$ axis and lie on the primary branch. In summary, the root-locus branch corresponding to $k = 0$ is the dominant one; other branches corresponding to $k = 1, 2, 3, \dots$ are not so important and may be neglected.

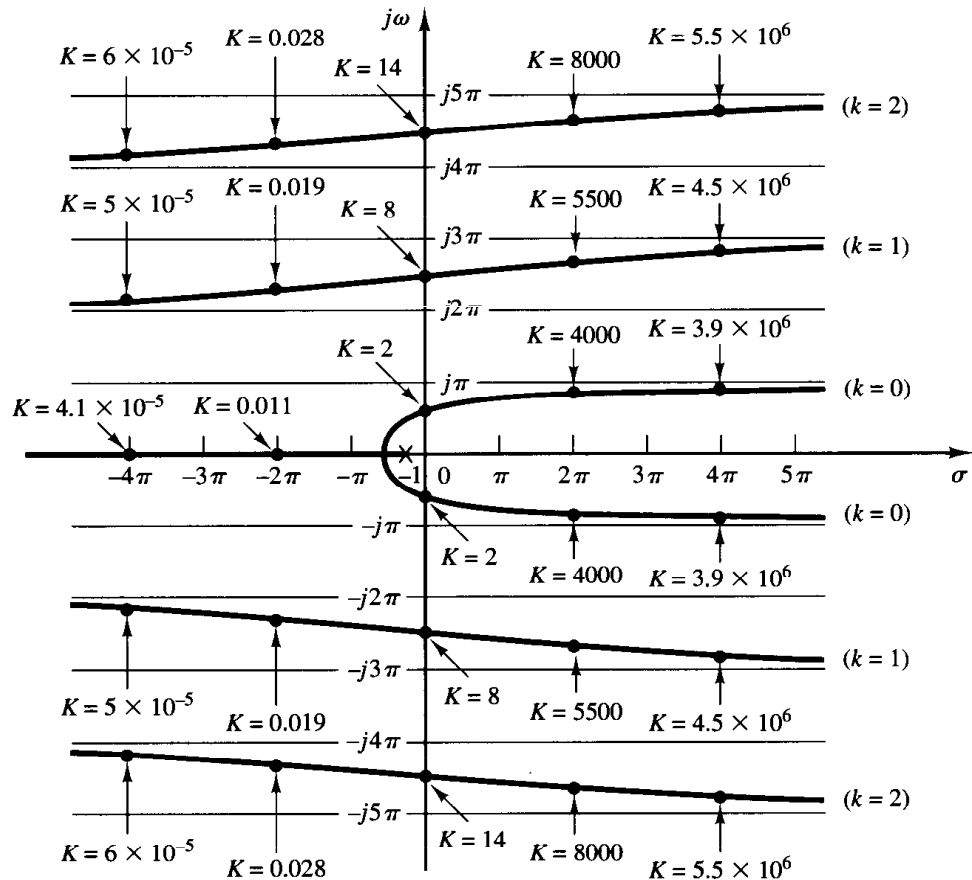


Figure 6–38
Root-locus plot for
the system shown
in Figure 6–36
($T = 1$ sec).

This example illustrates the fact that dead time can cause instability even in the first-order system because the root loci enter the right-half s plane for large values of K . Therefore, although the gain K of the first-order system can be set at a high value in the absence of dead time, it cannot be set too high if dead time is present. (For the system considered here, the value of gain K must be considerably less than 2 for a satisfactory operation.)

Approximation of transport lag or dead time. If the dead time T is very small, then e^{-Ts} is frequently approximated by

$$e^{-Ts} \doteq 1 - Ts$$

or

$$e^{-Ts} \doteq \frac{1}{Ts + 1}$$

Such approximations are good if the dead time is very small and, in addition, the input time function $f(t)$ to the dead-time element is smooth and continuous. [This means that the second- and higher-order derivatives of $f(t)$ are small.]

A more elaborate expression to approximate e^{-Ts} is available and is

$$e^{-Ts} = \frac{1 - \frac{Ts}{2} + \frac{(Ts)^2}{8} - \frac{(Ts)^3}{48} + \dots}{1 + \frac{Ts}{2} + \frac{(Ts)^2}{8} + \frac{(Ts)^3}{48} + \dots}$$

If only the first two terms in the numerator and denominator are taken, then

$$e^{-Ts} \doteq \frac{1 - \frac{Ts}{2}}{1 + \frac{Ts}{2}} = \frac{2 - Ts}{2 + Ts}$$

This approximation is also used frequently.

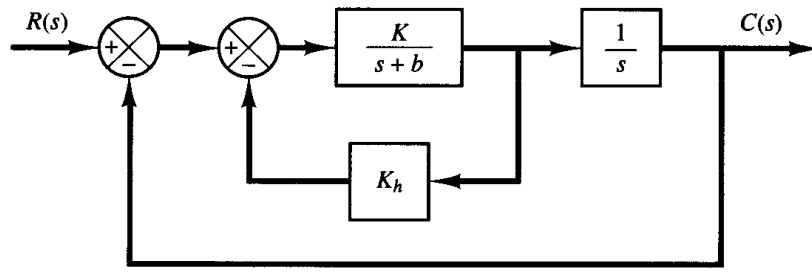
6-8 ROOT-CONTOUR PLOTS

Effects of parameter variations on closed-loop poles. In many design problems, the effects on the closed-loop poles of the variations of parameters other than the gain K need to be investigated. Such effects can be easily investigated by the root-locus method. When two (or more) parameters are varied, the corresponding root loci are called *root contours*.

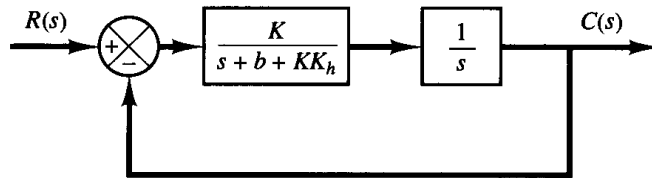
We shall use an example to illustrate the construction of the root contours when two parameters are varied, respectively, from zero to infinity.

Consider a servo system having tachometer feedback as shown in Figure 6-39(a). By eliminating the minor loop, the block diagram can be simplified [Figure 6-39(b)]. By defining

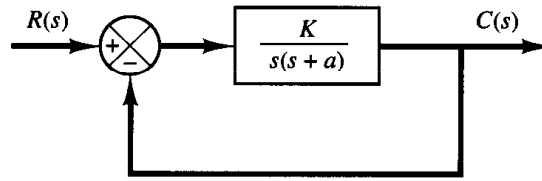
$$a = b + KK_h$$



(a)



(b)



(c)

Figure 6–39
(a) Servo system with tachometer feedback; (b), (c) simplified block diagrams ($a = b + KK_h$).

this block diagram can be modified to that shown in Figure 6–39(c). This system involves two variables, parameter a and gain K .

In what follows we shall investigate the effect of varying the parameter a as well as the gain K . The closed-loop transfer function of this system becomes

$$\frac{C(s)}{R(s)} = \frac{K}{s^2 + as + K}$$

The characteristic equation is

$$s^2 + as + K = 0 \quad (6-27)$$

which may be rewritten

$$1 + \frac{as}{s^2 + K} = 0$$

or

$$\frac{as}{s^2 + K} = -1 \quad (6-28)$$

In Equation (6–28), the parameter a is written as a multiplying factor. For a given value of K , the effect of a on the closed-loop poles can be investigated from Equation (6–28). The root contours for this system can be constructed by following the usual procedure for constructing root loci.

We shall now construct the root contours as K and a vary, respectively, from zero to infinity. The root contours start from the poles (at $s = \pm j\sqrt{K}$) and terminate at the zeros (at $s = 0$ and infinity).

We shall first construct the locus of roots when $a = 0$. This can be done easily as follows: Substitute $a = 0$ into Equation (6-27). Then

$$s^2 + K = 0$$

or

$$\frac{K}{s^2} = -1 \quad (6-29)$$

The open-loop poles are thus a double pole at the origin. The root-locus plot of Equation (6-29) is shown in Figure 6-40(a).

To construct the root contours, let us assume that K is a constant; for example, $K = 4$. Then Equation (6-28) becomes

$$\frac{as}{s^2 + 4} = -1 \quad (6-30)$$

The open-loop poles are $s = \pm j2$. The finite open-loop zero is at the origin. The root-locus plot corresponding to Equation (6-30) is shown in Figure 6-40(b). For different values of K , Equation (6-30) yields similar root loci.

The root contour, the diagram showing the root loci corresponding to $0 \leq K \leq \infty$, $0 \leq a \leq \infty$, can be plotted as in Figure 6-40(c). Clearly, the root contours start at the

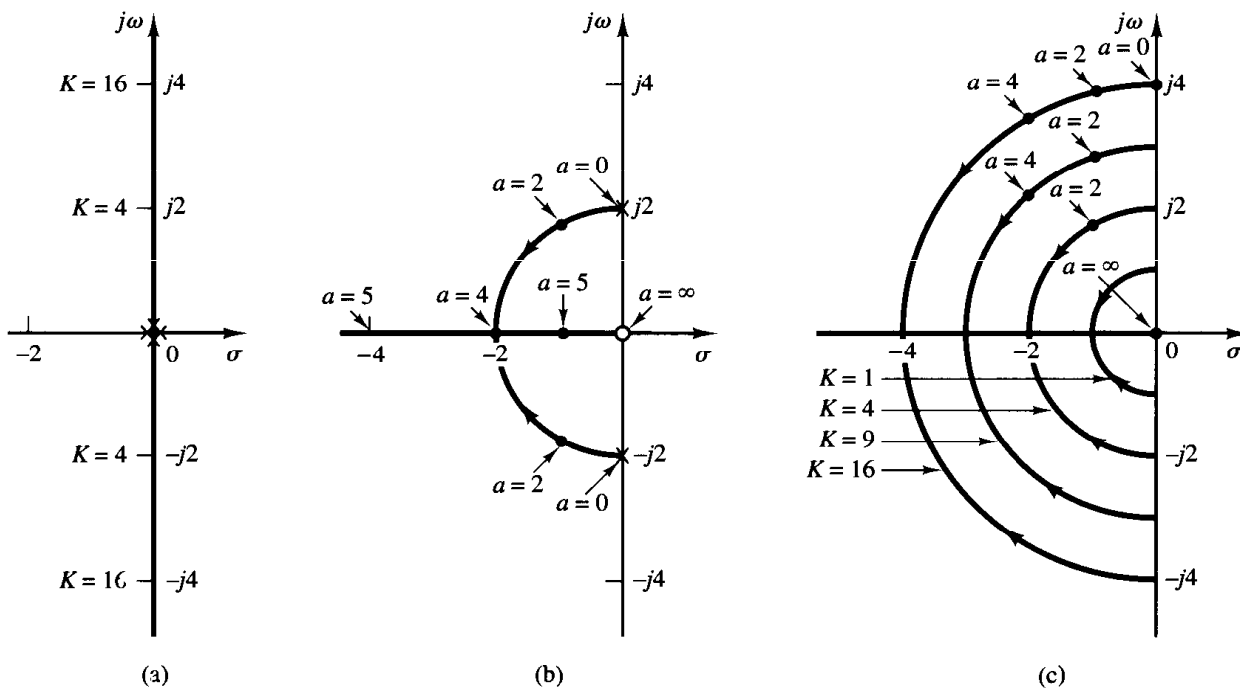


Figure 6-40

(a) Root-locus plot for the system shown in Figure 6-39(c) ($a = 0, 0 \leq K \leq \infty$); (b) root locus plot ($0 \leq a \leq \infty, K = 4$); (c) root-contour plot.

poles of and end at the zeros of the transfer function $as/(s^2 + K)$. The arrowheads on the root contours indicate the direction of increase in the value of a .

The root contours show the effects of the variations of system parameters on the closed-loop poles. From the root-contour plot shown in Figure 6-40(c), we see that, for $0 < K < \infty$, $0 < a < \infty$, the closed-loop poles lie in the left-half s plane and the system is stable.

Note that if the value of K is fixed, say $K = 4$, then the root contours become simply the root loci, as shown in Figure 6-40(b).

We have illustrated a method for constructing root contours when the gain K and parameter a are varied, respectively, from zero to infinity. Basically, we assign one parameter a constant value at a time and vary the other parameter from 0 to ∞ and sketch the root loci. Then we change the value of the first parameter and repeat sketching the root loci. By repeating this process we can sketch the root contour.

A MATLAB program to generate the root-contour plot is given in MATLAB Program 6-9. The resulting plot is shown in Figure 6-41.

MATLAB Program 6-9

```
% ----- Root-contour plot -----

% ***** Plot root contour of the system shown in Figure 6-39(c),
% where a and K are variables *****

% ***** In Equation (6-28),  $as/(s^2 + K) = -1$ , assume  $K = 1, 4, 9,$ 
%  $16, \dots$  and plot root loci as  $a$  varies from zero to infinity *****

% ***** Enter the numerator and denominators *****

num = [0 1 0];
den1 = [1 0 1];
den2 = [1 0 4];
den3 = [1 0 9];
den4 = [1 0 16];

% ***** Enter rlocus(num,den) command *****

rlocus(num,den1)
hold
Current plot held
rlocus(num,den2)
rlocus(num,den3)
rlocus(num,den4)
v = [-5 2 -5 5]; axis(v); axis('square');
grid
title('Root-Contour Plot')

% ***** Remove hold on graphics *****

hold
Current plot released
```

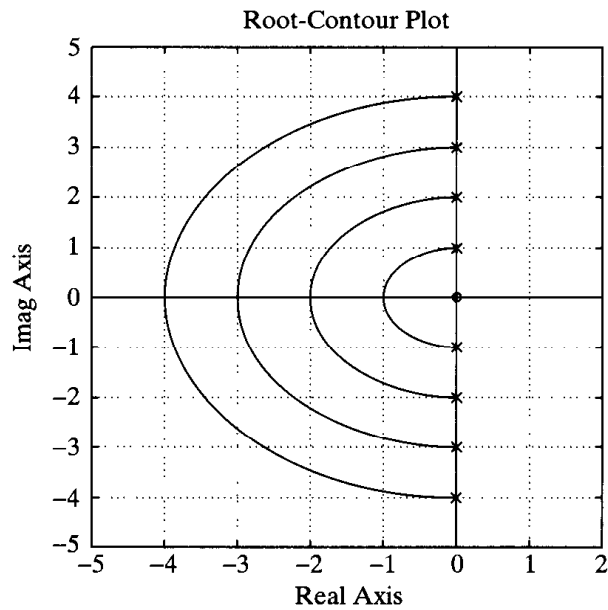


Figure 6-41
Root-contour plot
generated with
MATLAB.

EXAMPLE PROBLEMS AND SOLUTIONS

- A-6-1.** Sketch the root loci for the system shown in Figure 6-42(a). (The gain K is assumed to be positive.) Observe that for small or large values of K the system is overdamped and for medium values of K it is underdamped.

Solution. The procedure for plotting the root loci is as follows:

1. Locate the open-loop poles and zeros on the complex plane. Root loci exist on the negative real axis between 0 and -1 and between -2 and -3 .

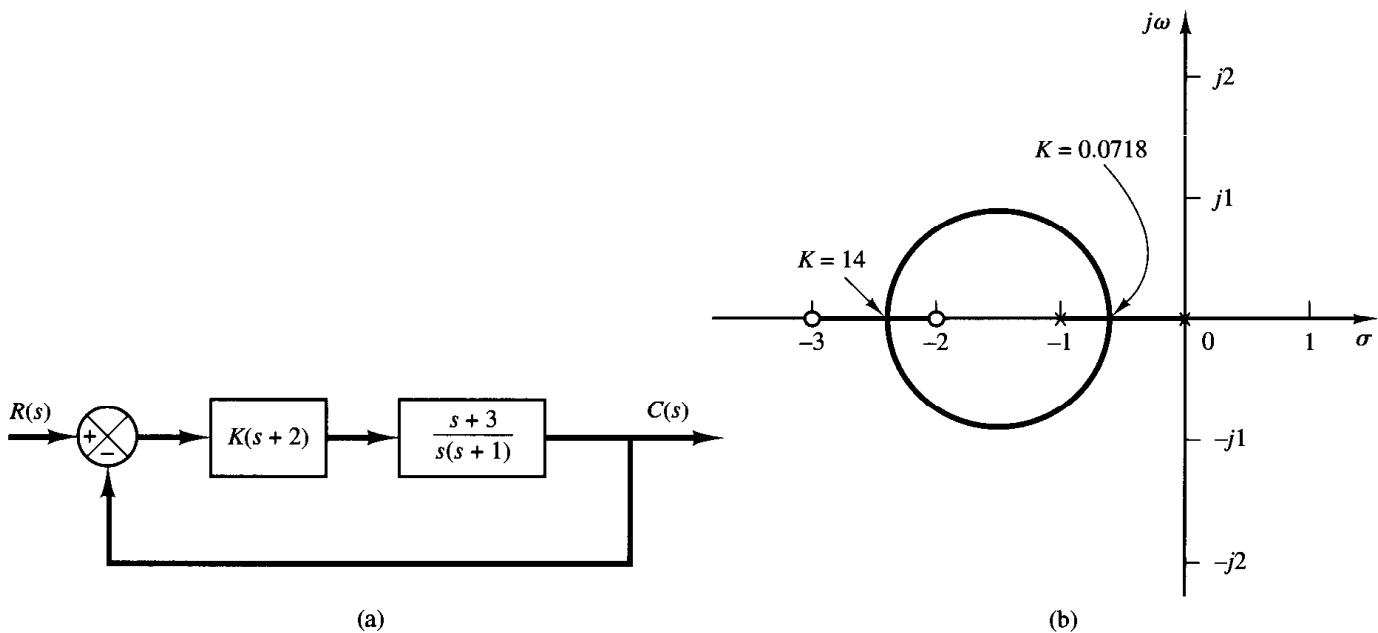


Figure 6-42
(a) Control system; (b) root-locus plot.

2. The number of open-loop poles and that of finite zeros are the same. This means that there are no asymptotes in the complex region of the s plane.
3. Determine the breakaway and break-in points. The characteristic equation for the system is

$$1 + \frac{K(s+2)(s+3)}{s(s+1)} = 0$$

or

$$K = -\frac{s(s+1)}{(s+2)(s+3)}$$

The breakaway and break-in points are determined from

$$\begin{aligned} \frac{dK}{ds} &= -\frac{(2s+1)(s+2)(s+3) - s(s+1)(2s+5)}{[(s+2)(s+3)]^2} \\ &= -\frac{4(s+0.634)(s+2.366)}{[(s+2)(s+3)]^2} \\ &= 0 \end{aligned}$$

as follows:

$$s = -0.634, \quad s = -2.366$$

Notice that both points are on root loci. Therefore, they are actual breakaway or break-in points. At point $s = -0.634$, the value of K is

$$K = -\frac{(-0.634)(0.366)}{(1.366)(2.366)} = 0.0718$$

Similarly, at $s = -2.366$,

$$K = -\frac{(-2.366)(-1.366)}{(-0.366)(0.634)} = 14$$

(Because point $s = -0.634$ lies between two poles, it is a breakaway point and because point $s = -2.366$ lies between two zeros, it is a break-in point.)

4. Determine a sufficient number of points that satisfy the angle condition. (It can be found that the root locus is a circle with center at -1.5 that passes through the breakaway and break-in points.) The root-locus plot for this system is shown in Figure 6-42(b).

Note that this system is stable for any positive value of K since all the root loci lie in the left-half s plane.

Small values of K ($0 < K < 0.0718$) correspond to an overdamped system. Medium values of K ($0.0718 < K < 14$) correspond to an underdamped system. Finally, large values of K ($14 < K$) correspond to an overdamped system. With a large value of K , the steady state can be reached in much shorter time than with a small value of K .

The value of K should be adjusted so that system performance is optimum according to a given performance index.

- A-6-2.** A simplified form of the open-loop transfer function of an airplane with an autopilot in the longitudinal mode is

$$G(s)H(s) = \frac{K(s+a)}{s(s-b)(s^2 + 2\zeta\omega_n s + \omega_n^2)}, \quad a > 0, \quad b > 0$$

Such a system involving an open-loop pole in the right-half s plane may be conditionally stable. Sketch the root loci when $a = b = 1$, $\zeta = 0.5$, and $\omega_n = 4$. Find the range of gain K for stability.

Solution. The open-loop transfer function for the system is

$$G(s)H(s) = \frac{K(s + 1)}{s(s - 1)(s^2 + 4s + 16)}$$

To sketch the root loci, we follow this procedure:

1. Locate the open-loop poles and zero in the complex plane. Root loci exist on the real axis between 1 and 0 and between -1 and $-\infty$.
2. Determine the asymptotes of the root loci. There are three asymptotes whose angles can be determined as

$$\text{Angles of asymptotes} = \frac{180^\circ(2k + 1)}{4 - 1} = 60^\circ, -60^\circ, 180^\circ$$

Referring to Equation (6-15), the abscissa of the intersection of the asymptotes and the real axis is

$$\sigma_a = -\frac{(0 - 1 + 2 + j2\sqrt{3} + 2 - j2\sqrt{3}) - 1}{4 - 1} = -\frac{2}{3}$$

3. Determine the breakaway and break-in points. Since the characteristic equation is

$$1 + \frac{K(s + 1)}{s(s - 1)(s^2 + 4s + 16)} = 0$$

we obtain

$$K = -\frac{s(s - 1)(s^2 + 4s + 16)}{s + 1}$$

By differentiating K with respect to s , we get

$$\frac{dK}{ds} = -\frac{3s^4 + 10s^3 + 21s^2 + 24s - 16}{(s + 1)^2}$$

The numerator can be factored as follows:

$$\begin{aligned} 3s^4 + 10s^3 + 21s^2 + 24s - 16 \\ = 3(s + 0.76 + j2.16)(s + 0.76 - j2.16)(s + 2.26)(s - 0.45) \end{aligned}$$

Points $s = 0.45$ and $s = -2.26$ are on root loci on the real axis. Hence, these points are actual breakaway and break-in points, respectively. Points $s = -0.76 \pm j2.16$ do not satisfy the angle condition. Hence, they are neither breakaway nor break-in points.

4. Using Routh's stability criterion, determine the value of K at which the root loci cross the imaginary axis. Since the characteristic equation is

$$s^4 + 3s^3 + 12s^2 + (K - 16)s + K = 0$$

the Routh array becomes

s^4	1	12	K
s^3	3	$K - 16$	0
s^2	$\frac{52 - K}{3}$	K	0
s^1	$\frac{-K^2 + 59K - 832}{52 - K}$	0	
s^0	K		

The values of K that make the s^1 term in the first column equal zero are $K = 35.7$ and $K = 23.3$.

The crossing points on the imaginary axis can be found by solving the auxiliary equation obtained from the s^2 row, that is, by solving the following equation for s :

$$\frac{52 - K}{3} s^2 + K = 0$$

The results are

$$s = \pm j2.56, \quad \text{for } K = 35.7$$

$$s = \pm j1.56, \quad \text{for } K = 23.3$$

The crossing points on the imaginary axis are thus $s = \pm j2.56$ and $s = \pm j1.56$.

5. Find the angles of departure of the root loci from the complex poles. For the open-loop pole at $s = -2 + j2\sqrt{3}$, the angle of departure θ is

$$\theta = 180^\circ - 120^\circ - 130.5^\circ - 90^\circ + 106^\circ$$

or

$$\theta = -54.5^\circ$$

(The angle of departure from the open-loop pole at $s = -2 - j2\sqrt{3}$ is 54.5° .)

6. Choose a test point in the broad neighborhood of the $j\omega$ axis and the origin and apply the angle condition. If the test point does not satisfy the angle condition, select another test point until it does. Continue the same process and locate a sufficient number of points that satisfy the angle condition.

Figure 6-43 shows the root loci for this system. From step 4 the system is stable for $23.3 < K < 35.7$. Otherwise, it is unstable.

A-6-3. Sketch the root loci of the control system shown in Figure 6-44(a).

Solution. The open-loop poles are located at $s = 0$, $s = -3 + j4$, and $s = -3 - j4$. A root locus branch exists on the real axis between the origin and $-\infty$. There are three asymptotes for the root loci. The angles of asymptotes are

$$\text{Angles of asymptotes} = \frac{\pm 180^\circ(2k + 1)}{3} = 60^\circ, -60^\circ, 180^\circ$$

Referring to Equation (6-15), the intersection of the asymptotes and the real axis is obtained as

$$\sigma_a = -\frac{0 + 3 + 3}{3} = -2$$

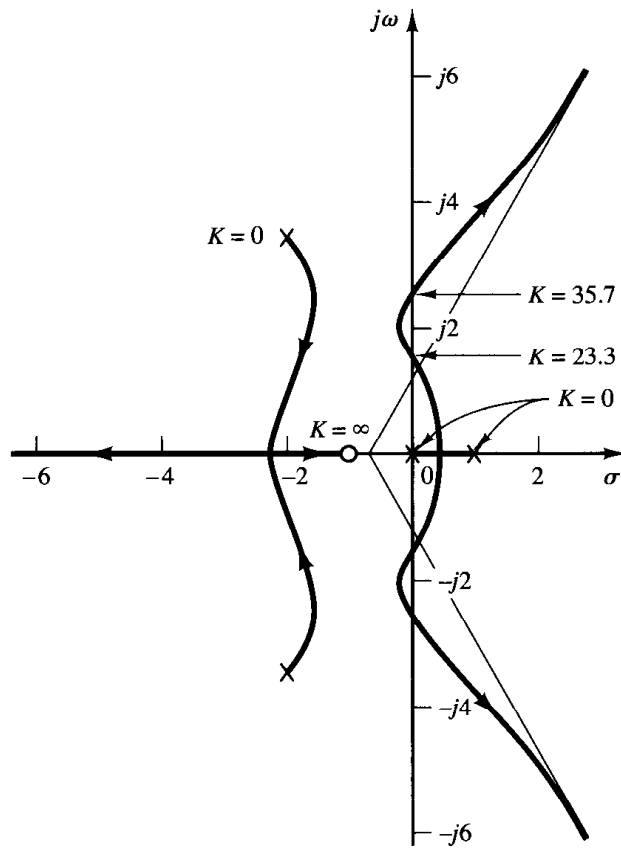


Figure 6-43
Root-locus plot.

Next we check the breakaway and break-in points. For this system we have

$$K = -s(s^2 + 6s + 25)$$

Now we set

$$\frac{dK}{ds} = -(3s^2 + 12s + 25) = 0$$

which yields

$$s = -2 + j2.0817, \quad s = -2 - j2.0817$$

Notice that at points $s = -2 \pm j2.0817$ the angle condition is not satisfied. Hence, they are neither breakaway nor break-in points. In fact, if we calculate the value of K , we obtain

$$K = -s(s^2 + 6s + 25) \Big|_{s=-2 \pm j2.0817} = 34 \pm j18.04$$

(To be an actual breakaway or break-in point, the corresponding value of K must be real and positive.)

The angle of departure from the complex pole in the upper half s plane is

$$\theta = 180^\circ - 126.87^\circ - 90^\circ$$

or

$$\theta = -36.87^\circ$$

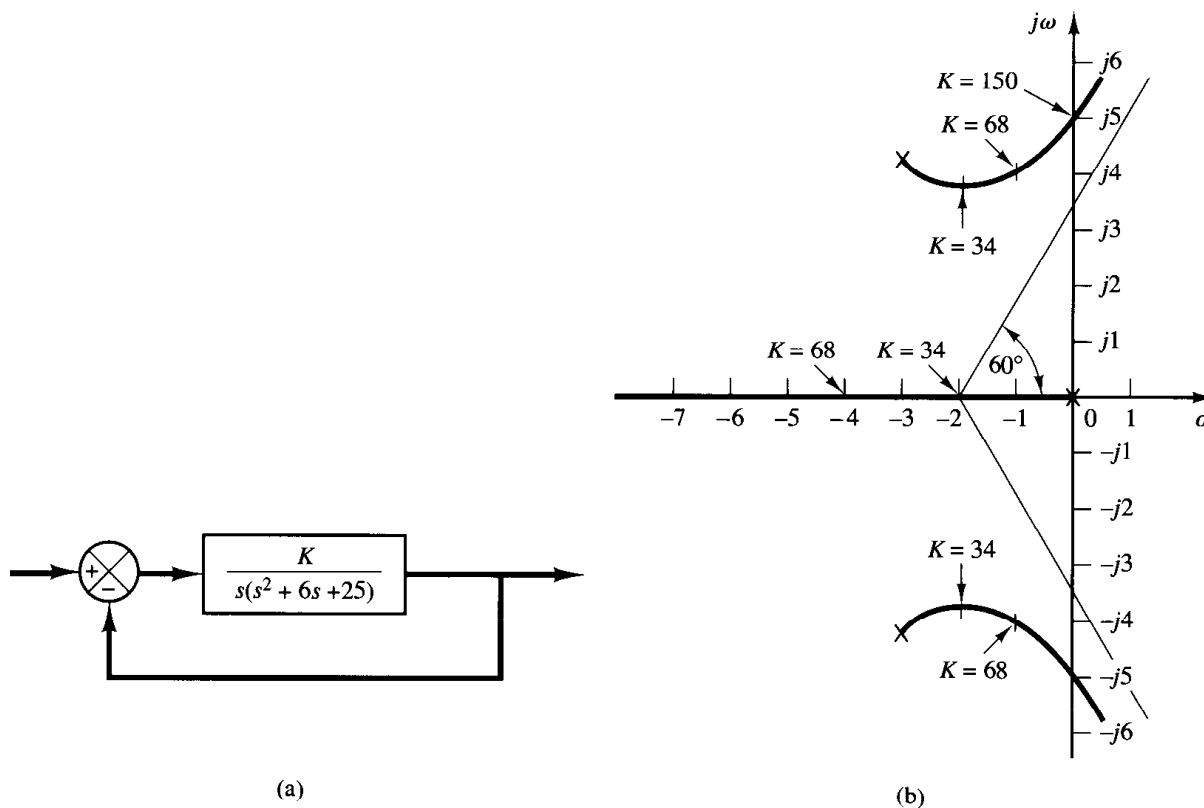


Figure 6-44

(a) Control system; (b) root-locus plot.

The points where root-locus branches cross the imaginary axis may be found by substituting $s = j\omega$ into the characteristic equation and solving the equation for ω and K as follows: Noting that the characteristic equation is

$$s^3 + 6s^2 + 25s + K = 0$$

we have

$$(j\omega)^3 + 6(j\omega)^2 + 25(j\omega) + K = (-6\omega^2 + K) + j\omega(25 - \omega^2) = 0$$

which yields

$$\omega = \pm 5, \quad K = 150 \quad \text{or} \quad \omega = 0, \quad K = 0$$

Root-locus branches cross the imaginary axis at $\omega = 5$ and $\omega = -5$. The value of gain K at the crossing points is 150. Also, the root-locus branch on the real axis touches the imaginary axis at $\omega = 0$. Figure 6-44(b) shows a root-locus plot for the system.

It is noted that if the order of the numerator of $G(s)H(s)$ is lower than that of the denominator by two or more, and if some of the closed-loop poles move on the root locus toward the right as gain K is increased, then other closed-loop poles must move toward the left as gain K is increased. This fact can be seen clearly in this problem. If the gain K is increased from $K = 34$ to $K = 68$, the complex-conjugate closed-loop poles are moved from $s = -2 + j3.65$ to $s = -1 + j4$; the third pole is moved from $s = -2$ (which corresponds to $K = 34$) to $s = -4$ (which corresponds to $K = 68$). Thus, the movements of two complex-conjugate closed-loop poles to the right by one unit cause the remaining closed-loop pole (real pole in this case) to move to the left by two units.

A-6-4. Consider the system shown in Figure 6–45(a). Sketch the root loci for the system. Observe that for small or large values of K the system is underdamped and for medium values of K it is overdamped.

Solution. A root locus exists on the real axis between the origin and $-\infty$. The angles of asymptotes of the root-locus branches are obtained as

$$\text{Angles of asymptotes} = \frac{\pm 180^\circ(2k + 1)}{3} = 60^\circ, -60^\circ, -180^\circ$$

The intersection of the asymptotes and the real axis is located on the real axis at

$$\sigma_a = -\frac{0 + 2 + 2}{3} = -1.3333$$

The breakaway and break-in points are found from $dK/ds = 0$. Since the characteristic equation is

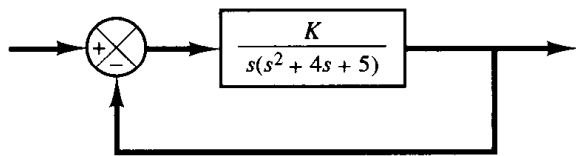
$$s^3 + 4s^2 + 5s + K = 0$$

we have

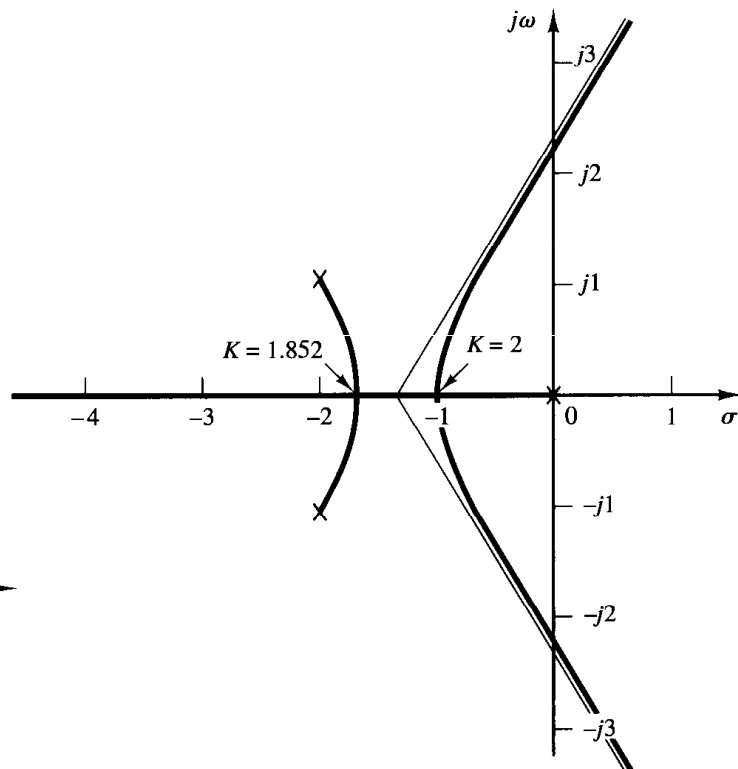
$$K = -(s^3 + 4s^2 + 5s)$$

Now we set

$$\frac{dK}{ds} = -(3s^2 + 8s + 5) = 0$$



(a)



(b)

Figure 6–45
(a) Control system; (b) root-locus plot.

which yields

$$s = -1, \quad s = -1.6667$$

Since these points are on root loci, they are actual breakaway or break-in points. (At point $s = -1$, the value of K is 2, and at point $s = -1.6667$, the value of K is 1.852.)

The angle of departure from a complex pole in the upper half s plane is obtained from

$$\theta = 180^\circ - 153.43^\circ - 90^\circ$$

or

$$\theta = -63.43^\circ$$

The root-locus branch from the complex pole in the upper half s plane breaks into the real axis at $s = -1.6667$.

Next we determine the points where root-locus branches cross the imaginary axis. By substituting $s = j\omega$ into the characteristic equation, we have

$$(j\omega)^3 + 4(j\omega)^2 + 5(j\omega) + K = 0$$

or

$$(K - 4\omega^2) + j\omega(5 - \omega^2) = 0$$

from which we obtain

$$\omega = \pm\sqrt{5}, \quad K = 20 \quad \text{or} \quad \omega = 0, \quad K = 0$$

Root-locus branches cross the imaginary axis at $\omega = \sqrt{5}$ and $\omega = -\sqrt{5}$. The root-locus branch on the real axis touches the $j\omega$ axis at $\omega = 0$. A sketch of the root loci for the system is shown in Figure 6-45(b).

Note that since this system is of third order there are three closed-loop poles. The nature of the system response to a given input depends on the locations of the closed-loop poles.

For $0 < K < 1.852$, there are a set of complex-conjugate closed-loop poles and a real closed-loop pole. For $1.852 \leq K \leq 2$, there are three real closed-loop poles. For example, the closed-loop poles are located at

$$\begin{array}{llll} s = -1.667, & s = -1.667, & s = -0.667, & \text{for } K = 1.852 \\ s = -1, & s = -1, & s = -2, & \text{for } K = 2 \end{array}$$

For $2 < K$, there are a set of complex-conjugate closed-loop poles and a real closed-loop pole. Thus, small values of K ($0 < K < 1.852$) correspond to an underdamped system. (Since the real closed-loop pole dominates, only a small ripple may show up in the transient response.) Medium values of K ($1.852 \leq K \leq 2$) correspond to an overdamped system. Large values of K ($2 < K$) correspond to an underdamped system. With a large value of K , the system responds much faster than with a smaller value of K .

A-6-5. Sketch the root loci for the system shown in Figure 6-46(a).

Solution. The open-loop poles are located at $s = 0$, $s = -1$, $s = -2 + j3$, and $s = -2 - j3$. A root locus exists on the real axis between points $s = 0$ and $s = -1$. The asymptotes are found as follows:

$$\text{Angles of asymptotes} = \frac{\pm 180^\circ(2k + 1)}{4} = 45^\circ, -45^\circ, 135^\circ, -135^\circ$$

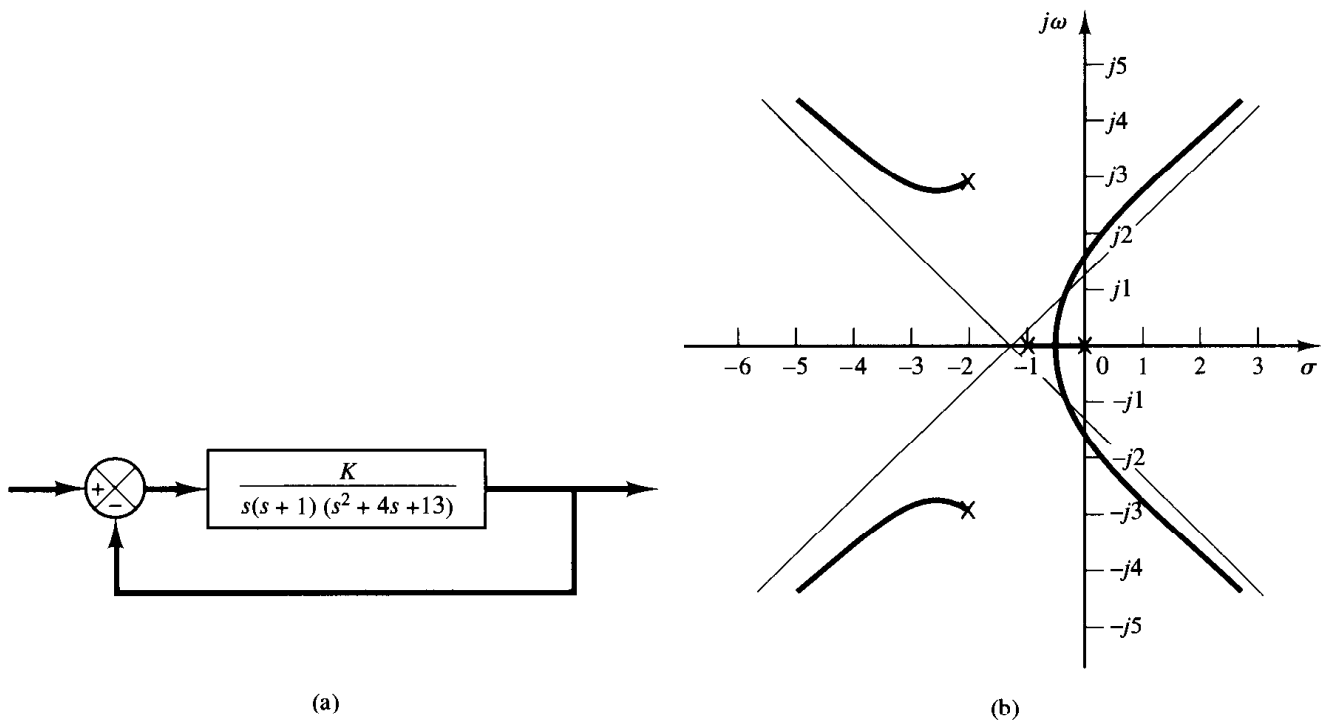


Figure 6-46

(a) Control system; (b) root-locus plot.

The intersection of the asymptotes and the real axis is found from

$$\sigma_a = -\frac{0 + 1 + 2 + 2}{4} = -1.25$$

The breakaway and break-in points are found from $dK/ds = 0$. Noting that

$$K = -s(s+1)(s^2+4s+13) = -(s^4+5s^3+17s^2+13s)$$

we have

$$\frac{dK}{ds} = -(4s^3 + 15s^2 + 34s + 13) = 0$$

from which we get

$$s = -0.467, \quad s = -1.642 + j2.067, \quad s = -1.642 - j2.067$$

The point $s = -0.467$ is on a root locus. Therefore, it is an actual breakaway point. The gain values K corresponding to points $s = -1.642 \pm j2.067$ are complex quantities. Since the gain values are not real positive, these points are neither breakaway nor break-in points.

The angle of departure from the complex pole in the upper half s plane is

$$\theta = 180^\circ - 123.69^\circ - 108.44^\circ - 90^\circ$$

or

$$\theta = -142.13^\circ$$

Next we shall find the points where root loci may cross the $j\omega$ axis. Since the characteristic equation is

$$s^4 + 5s^3 + 17s^2 + 13s + K = 0$$

by substituting $s = j\omega$ into it we obtain

$$(j\omega)^4 + 5(j\omega)^3 + 17(j\omega)^2 + 13(j\omega) + K = 0$$

or

$$(K + \omega^4 - 17\omega^2) + j\omega(13 - 5\omega^2) = 0$$

from which we obtain

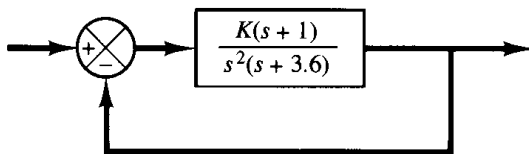
$$\omega = \pm 1.6125, \quad K = 37.44 \quad \text{or} \quad \omega = 0, \quad K = 0$$

The root-locus branches that extend to the right-half s plane cross the imaginary axis at $\omega = \pm 1.6125$. Also, the root-locus branch on the real axis touches the imaginary axis at $\omega = 0$. Figure 6–46(b) shows a sketch of the root loci for the system. Notice that each root-locus branch that extends to the right half s plane crosses its own asymptote.

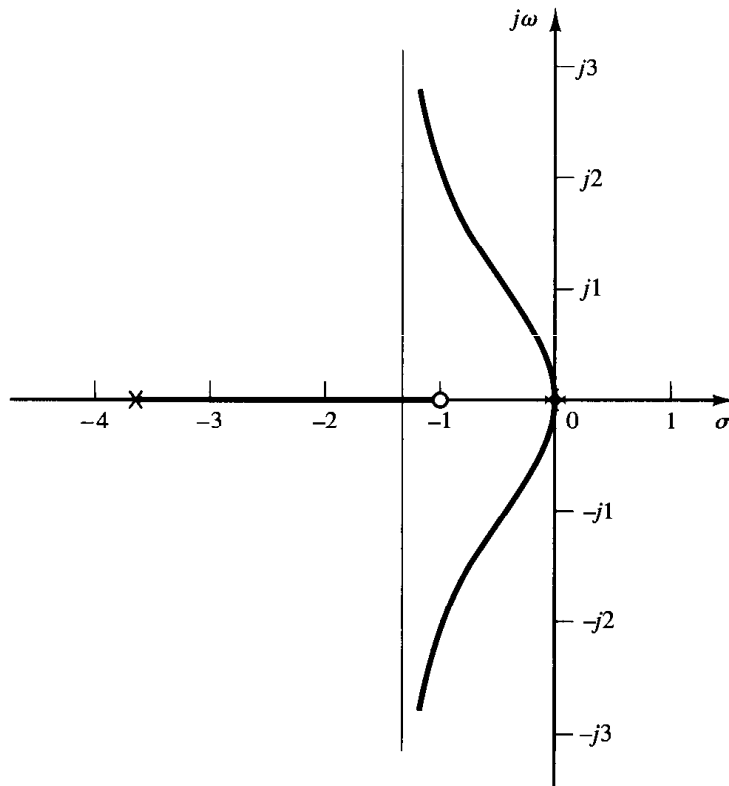
A-6-6. Sketch the root loci for the system shown in Figure 6–47(a).

Solution. A root locus exists on the real axis between points $s = -1$ and $s = -3.6$. The asymptotes can be determined as follows:

$$\text{Angles of asymptotes} = \frac{\pm 180^\circ(2k + 1)}{3 - 1} = 90^\circ, -90^\circ$$



(a)



(b)

Figure 6–47

(a) Control system; (b) root-locus plot.

The intersection of the asymptotes and the real axis is found from

$$\sigma_a = -\frac{0 + 0 + 3.6 - 1}{3 - 1} = -1.3$$

Since the characteristic equation is

$$s^3 + 3.6s^2 + K(s + 1) = 0$$

we have

$$K = -\frac{s^3 + 3.6s^2}{s + 1}$$

The breakaway and break-in points are found from

$$\frac{dK}{ds} = -\frac{(3s^2 + 7.2s)(s + 1) - (s^3 + 3.6s^2)}{(s + 1)^2} = 0$$

or

$$s^3 + 3.3s^2 + 3.6s = 0$$

from which we get

$$s = 0, \quad s = -1.65 + j0.9367, \quad s = -1.65 - j0.9367$$

Point $s = 0$ corresponds to the actual breakaway point. But points $s = -1.65 \pm j0.9367$ are neither breakaway nor break-in points, because the corresponding gain values K become complex quantities.

To check the points where root-locus branches may cross the imaginary axis, substitute $s = j\omega$ into the characteristic equation.

$$(j\omega)^3 + 3.6(j\omega)^2 + Kj\omega + K = 0$$

or

$$(K - 3.6\omega^2) + j\omega(K - \omega^2) = 0$$

Notice that this equation can be satisfied only if $\omega = 0$, $K = 0$. Because of the presence of a double pole at the origin, the root locus is tangent to the $j\omega$ axis at $\omega = 0$. The root-locus branches do not cross the $j\omega$ axis. Figure 6-47(b) is a sketch of the root loci for this system.

A-6-7. Sketch the root loci for the system shown in Figure 6-48(a).

Solution. A root locus exists on the real axis between point $s = -0.4$ and $s = -3.6$. The asymptotes can be found as follows:

$$\text{Angles of asymptotes} = \frac{\pm 180^\circ(2k + 1)}{3 - 1} = 90^\circ, -90^\circ$$

The intersection of the asymptotes and the real axis is obtained from

$$\sigma_a = -\frac{0 + 0 + 3.6 - 0.4}{3 - 1} = -1.6$$

Next we shall find the breakaway points. Since the characteristic equation is

$$s^3 + 3.6s^2 + Ks + 0.4K = 0$$

we have

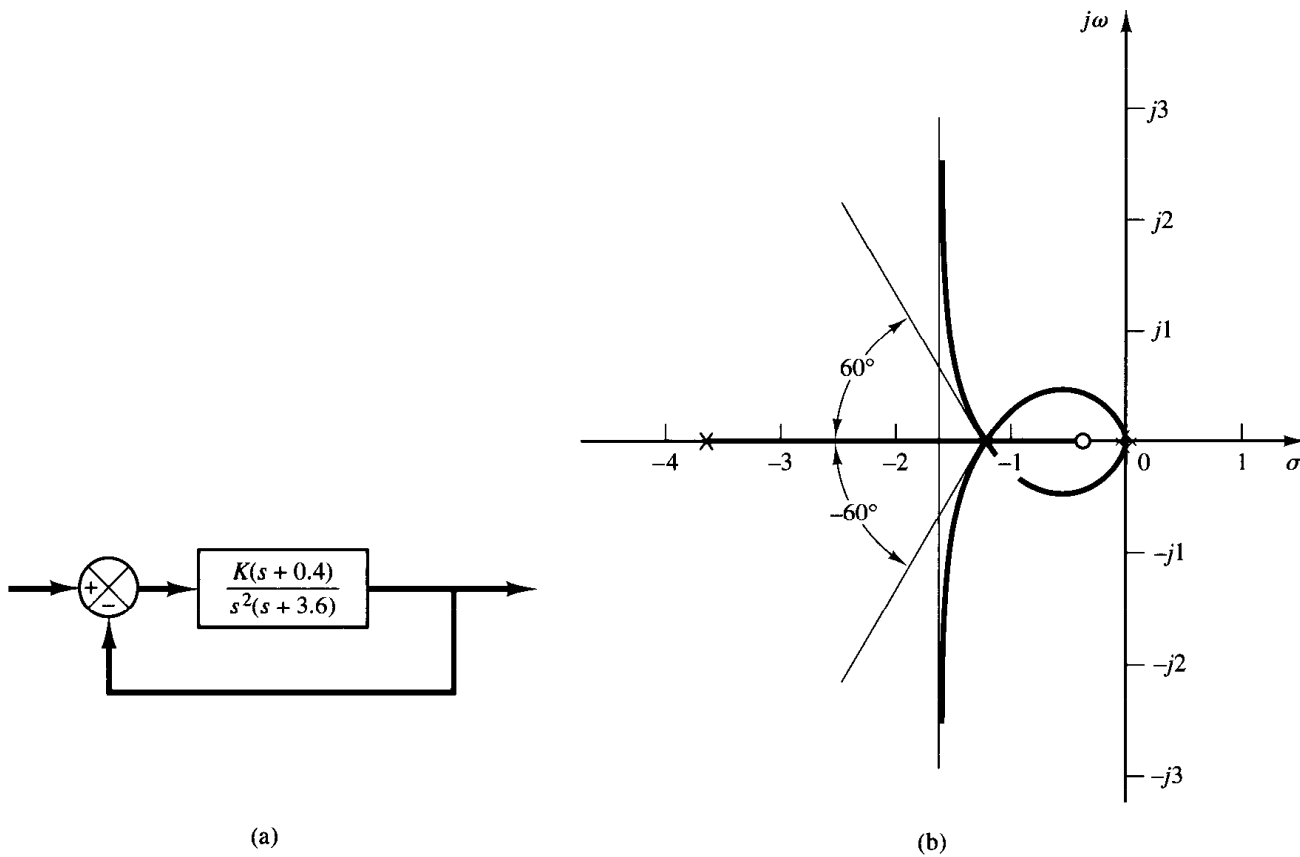


Figure 6-48
(a) Control system; (b) root-locus plot.

$$K = -\frac{s^3 + 3.6s^2}{s + 0.4}$$

The breakaway and break-in points are found from

$$\frac{dK}{ds} = -\frac{(3s^2 + 7.2s)(s + 0.4) - (s^3 + 3.6s^2)}{(s + 0.4)^2} = 0$$

from which we get

$$s^3 + 2.4s^2 + 1.44s = 0$$

or

$$s(s + 1.2)^2 = 0$$

Thus, the breakaway or break-in points are at $s = 0$ and $s = -1.2$. Note that $s = -1.2$ is a double root. When a double root occurs in $dK/ds = 0$ at point $s = -1.2$, $d^2K/(ds^2) = 0$ at this point. The value of gain K at point $s = -1.2$ is

$$K = -\frac{s^3 + 3.6s^2}{s + 0.4} \bigg|_{s=-1.2} = 4.32$$

This means that with $K = 4.32$ the characteristic equation has a triple root at point $s = -1.2$. This can be easily verified as follows:

$$s^3 + 3.6s^2 + 4.32s + 1.728 = (s + 1.2)^3 = 0$$

Hence, three root-locus branches meet at point $s = -1.2$. The angles of departures at point $s = -1.2$ of the root locus branches that approach the asymptotes are $\pm 180^\circ/3$, that is, 60° and -60° . (See Problem A-6-8.)

Finally, we shall examine if root-locus branches cross the imaginary axis. By substituting $s = j\omega$ into the characteristic equation, we have

$$(j\omega)^3 + 3.6(j\omega)^2 + K(j\omega) + 0.4K = 0$$

or

$$(0.4K - 3.6\omega^2) + j\omega(K - \omega^2) = 0$$

This equation can be satisfied only if $\omega = 0, K = 0$. At point $\omega = 0$, the root locus is tangent to the $j\omega$ axis because of the presence of a double pole at the origin. There are no points that root-locus branches cross the imaginary axis.

A sketch of the root loci for this system is shown in Figure 6-48(b).

- A-6-8.** Referring to Problem A-6-7, obtain the equations for the root-locus branches for the system shown in Figure 6-48(a). Show that the root-locus branches cross the real axis at the breakaway point at angles $\pm 60^\circ$.

Solution. The equations for the root-locus branches can be obtained from the angle condition

$$\angle \frac{K(s + 0.4)}{s^2(s + 3.6)} = \pm 180^\circ(2k + 1)$$

which can be rewritten as

$$\angle s + 0.4 - 2\angle s - \angle s + 3.6 = \pm 180^\circ(2k + 1)$$

By substituting $s = \sigma + j\omega$, we obtain

$$\angle \sigma + j\omega + 0.4 - 2\angle \sigma + j\omega - \angle \sigma + j\omega + 3.6 = \pm 180^\circ(2k + 1)$$

or

$$\tan^{-1}\left(\frac{\omega}{\sigma + 0.4}\right) - 2\tan^{-1}\left(\frac{\omega}{\sigma}\right) - \tan^{-1}\left(\frac{\omega}{\sigma + 3.6}\right) = \pm 180^\circ(2k + 1)$$

By rearranging, we have

$$\tan^{-1}\left(\frac{\omega}{\sigma + 0.4}\right) - \tan^{-1}\left(\frac{\omega}{\sigma}\right) = \tan^{-1}\left(\frac{\omega}{\sigma}\right) + \tan^{-1}\left(\frac{\omega}{\sigma + 3.6}\right) \pm 180^\circ(2k + 1)$$

Taking tangents of both sides of this last equation, and noting that

$$\tan\left[\tan^{-1}\left(\frac{\omega}{\sigma + 3.6}\right) \pm 180^\circ(2k + 1)\right] = \frac{\omega}{\sigma + 3.6}$$

we obtain

$$\frac{\frac{\omega}{\sigma + 0.4} - \frac{\omega}{\sigma}}{1 + \frac{\omega}{\sigma + 0.4} \frac{\omega}{\sigma}} = \frac{\frac{\omega}{\sigma} + \frac{\omega}{\sigma + 3.6}}{1 - \frac{\omega}{\sigma} \frac{\omega}{\sigma + 3.6}}$$

which can be simplified to

$$\frac{\omega\sigma - \omega(\sigma + 0.4)}{(\sigma + 0.4)\sigma + \omega^2} = \frac{\omega(\sigma + 3.6) + \omega\sigma}{\sigma(\sigma + 3.6) - \omega^2}$$

or

$$\omega(\sigma^3 + 2.4\sigma^2 + 1.44\sigma + 1.6\omega^2 + \sigma\omega^2) = 0$$

which can be further simplified to

$$\omega[\sigma(\sigma + 1.2)^2 + (\sigma + 1.6)\omega^2] = 0$$

For $\sigma \neq -1.6$, we may write this last equation as

$$\omega \left[\omega - (\sigma + 1.2) \sqrt{\frac{-\sigma}{\sigma + 1.6}} \right] \left[\omega + (\sigma + 1.2) \sqrt{\frac{-\sigma}{\sigma + 1.6}} \right] = 0$$

which gives the equations for the root-locus as follows:

$$\omega = 0$$

$$\omega = (\sigma + 1.2) \sqrt{\frac{-\sigma}{\sigma + 1.6}}$$

$$\omega = -(\sigma + 1.2) \sqrt{\frac{-\sigma}{\sigma + 1.6}}$$

The equation $\omega = 0$ represents the real axis. The root locus for $0 \leq K \leq \infty$ is between points $s = -0.4$ and $s = -3.6$. (The real axis other than this line segment and the origin $s = 0$ corresponds to the root locus for $-\infty \leq K < 0$.)

The equations

$$\omega = \pm(\sigma + 1.2) \sqrt{\frac{-\sigma}{\sigma + 1.6}} \quad (6-31)$$

represent the complex branches for $0 \leq K \leq \infty$. These two branches lie between $\sigma = -1.6$ and $\sigma = 0$. [See Figure 6-48(b).] The slopes of the complex root-locus branches at the breakaway point ($\sigma = -1.2$) can be found by evaluating $d\omega/d\sigma$ of Equation (6-31) at point $\sigma = -1.2$.

$$\left. \frac{d\omega}{d\sigma} \right|_{\sigma=-1.2} = \pm \left. \sqrt{\frac{-\sigma}{\sigma + 1.6}} \right|_{\sigma=-1.2} = \pm \sqrt{\frac{1.2}{0.4}} = \pm \sqrt{3}$$

Since $\tan^{-1} \sqrt{3} = 60^\circ$, the root-locus branches intersect the real axis with angles $\pm 60^\circ$.

- A-6-9.** Consider the system shown in Figure 6-49, which has an unstable feedforward transfer function. Sketch the root-locus plot and locate the closed-loop poles. Show that, although the closed-loop poles lie on the negative real axis and the system is not oscillatory, the unit-step response curve will exhibit overshoot.

Solution. The root-locus plot for this system is shown in Figure 6-50. The closed-loop poles are located at $s = -2$ and $s = -5$.

The closed-loop transfer function becomes

$$\frac{C(s)}{R(s)} = \frac{10(s + 1)}{s^2 + 7s + 10}$$

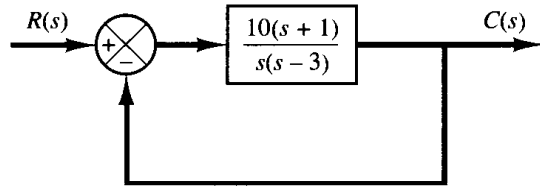


Figure 6-49
Control system.

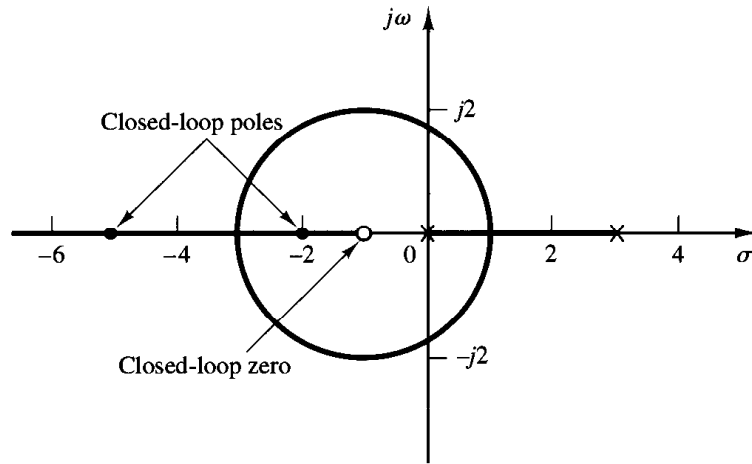


Figure 6-50
Root-locus plot for
the system shown in
Figure 6-49.

The unit-step response of this system is

$$C(s) = \frac{10(s+1)}{s(s+2)(s+5)}$$

The inverse Laplace transform of $C(s)$ gives

$$c(t) = 1 + 1.666e^{-2t} - 2.666e^{-5t}, \quad \text{for } t \geq 0$$

The unit-step response curve is shown in Figure 6-51. Although the system is not oscillatory, the unit-step response curve exhibits overshoot. (This is due to the presence of a zero at $s = -1$.)

A-6-10. Sketch the root loci of the control system shown in Figure 6-52(a). Determine the range of gain K for stability.

Solution. Open-loop poles are located at $s = 1$, $s = -2 + j\sqrt{3}$, and $s = -2 - j\sqrt{3}$. A root locus exists on the real axis between points $s = 1$ and $s = -\infty$. The asymptotes of the root-locus branches are found as follows:

$$\text{Angles of asymptotes} = \frac{\pm 180^\circ(2k+1)}{3} = 60^\circ, -60^\circ, 180^\circ$$

The intersection of the asymptotes and the real axis is obtained as

$$\sigma_a = -\frac{-1+2+2}{3} = -1$$

The breakaway and break-in points can be located from $dK/ds = 0$. Since

$$K = -(s-1)(s^2+4s+7) = -(s^3+3s^2+3s-7)$$

we have

$$\frac{dK}{ds} = -(3s^2+6s+3) = 0$$

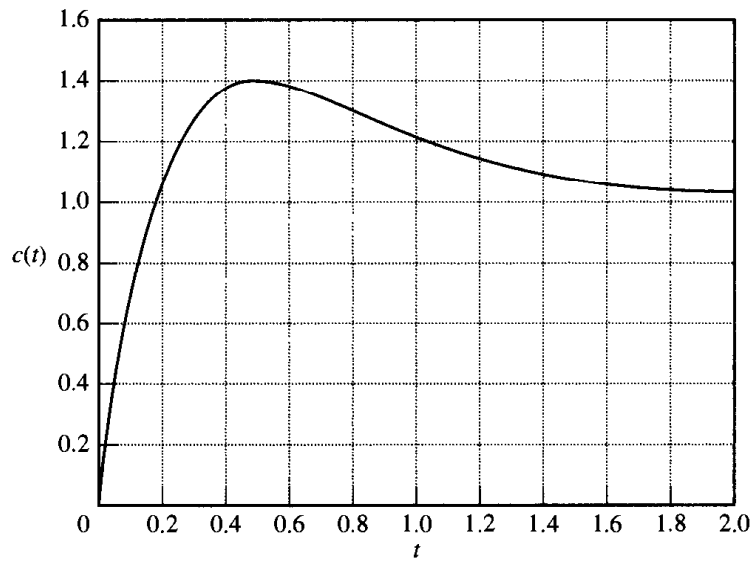


Figure 6-51
Unit-step response
curve for the system
shown in Figure
6-49.

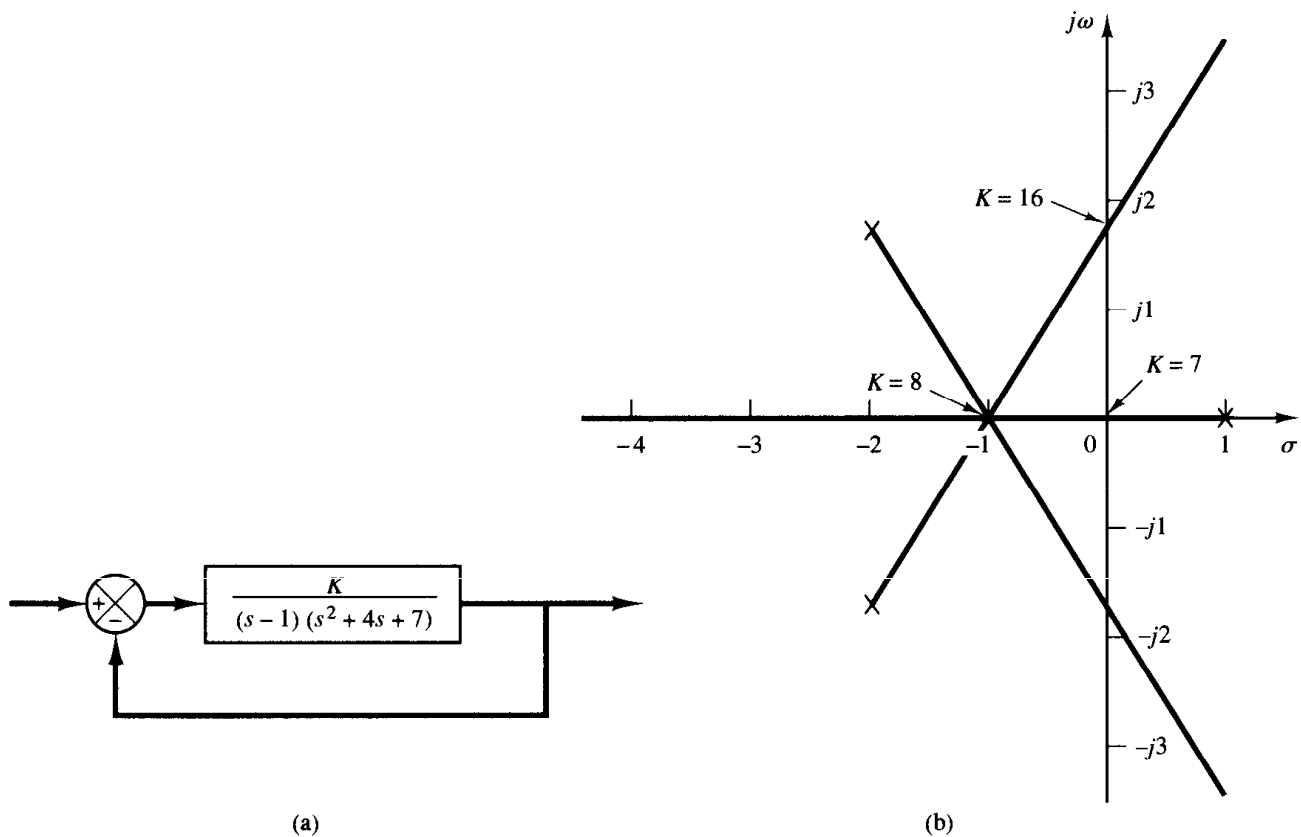


Figure 6-52
(a) Control system; (b) root-locus plot.

which yields

$$(s + 1)^2 = 0$$

Thus the equation $dK/ds = 0$ has a double root at $s = -1$. The breakaway point is located at $s = -1$. Three root locus branches meet at this breakaway point. The angles of departure of the branches at the breakaway point are $\pm 180^\circ/3$, that is, 60° and -60° .

We shall next determine the points where root-locus branches may cross the imaginary axis. Noting that the characteristic equation is

$$(s - 1)(s^2 + 4s + 7) + K = 0$$

or

$$s^3 + 3s^2 + 3s - 7 + K = 0$$

we substitute $s = j\omega$ into it and obtain

$$(j\omega)^3 + 3(j\omega)^2 + 3(j\omega) - 7 + K = 0$$

By rewriting this last equation, we have

$$(K - 7 - 3\omega^2) + j\omega(3 - \omega^2) = 0$$

This equation is satisfied when

$$\omega = \pm\sqrt{3}, \quad K = 7 + 3\omega^2 = 16 \quad \text{or} \quad \omega = 0, \quad K = 7$$

The root-locus branches cross the imaginary axis at $\omega = \pm\sqrt{3}$ (where $K = 16$) and $\omega = 0$ (where $K = 7$). Since the value of gain K at the origin is 7, the range of gain value K for stability is

$$7 < K < 16$$

Figure 6-52(b) shows a sketch of the root loci for the system. Notice that all branches consist of parts of straight lines.

The fact that the root-locus branches consist of straight lines can be verified as follows: Since the angle condition is

$$\angle \frac{K}{(s - 1)(s + 2 + j\sqrt{3})(s + 2 - j\sqrt{3})} = \pm 180^\circ(2k + 1)$$

we have

$$-\angle s - 1 - \angle s + 2 + j\sqrt{3} - \angle s + 2 - j\sqrt{3} = \pm 180^\circ(2k + 1)$$

By substituting $s = \sigma + j\omega$ into this last equation,

$$\angle \sigma - 1 + j\omega + \angle \sigma + 2 + j\omega + j\sqrt{3} + \angle \sigma + 2 + j\omega - j\sqrt{3} = \pm 180^\circ(2k + 1)$$

or

$$\angle \sigma + 2 + j(\omega + \sqrt{3}) + \angle \sigma + 2 + j(\omega - \sqrt{3}) = -\angle \sigma - 1 + j\omega \pm 180^\circ(2k + 1)$$

which can be rewritten as

$$\tan^{-1}\left(\frac{\omega + \sqrt{3}}{\sigma + 2}\right) + \tan^{-1}\left(\frac{\omega - \sqrt{3}}{\sigma + 2}\right) = -\tan^{-1}\left(\frac{\omega}{\sigma - 1}\right) \pm 180^\circ(2k + 1)$$

Taking tangents of both sides of this last equation, we obtain

$$\frac{\frac{\omega + \sqrt{3}}{\sigma + 2} + \frac{\omega - \sqrt{3}}{\sigma + 2}}{1 - \left(\frac{\omega + \sqrt{3}}{\sigma + 2}\right)\left(\frac{\omega - \sqrt{3}}{\sigma + 2}\right)} = -\frac{\omega}{\sigma - 1}$$

or

$$\frac{2\omega(\sigma + 2)}{\sigma^2 + 4\sigma + 4 - \omega^2 + 3} = -\frac{\omega}{\sigma - 1}$$

which can be simplified to

$$2\omega(\sigma + 2)(\sigma - 1) = -\omega(\sigma^2 + 4\sigma + 7 - \omega^2)$$

or

$$\omega(3\sigma^2 + 6\sigma + 3 - \omega^2) = 0$$

Further simplification of this last equation yields

$$\omega\left(\sigma + 1 + \frac{1}{\sqrt{3}}\omega\right)\left(\sigma + 1 - \frac{1}{\sqrt{3}}\omega\right) = 0$$

which defines three lines:

$$\omega = 0, \quad \sigma + 1 + \frac{1}{\sqrt{3}}\omega = 0, \quad \sigma + 1 - \frac{1}{\sqrt{3}}\omega = 0$$

Thus the root-locus branches consist of three lines. Note that the root loci for $K > 0$ consist of portions of the straight lines as shown in Figure 6-52(b). (Note that each straight line starts from an open-loop pole and extends to infinity in the direction of 180° , 60° , or -60° measured from the real axis.) The remaining portion of each straight line corresponds to $K < 0$.

A-6-11. Consider the system shown in Figure 6-53(a). Sketch the root loci.

Solution. The open-loop zeros of the system are located at $s = \pm j$. The open-loop poles are located at $s = 0$ and $s = -2$. This system involves two poles and two zeros. Hence, there is a possibility that a circular root-locus branch exists. In fact, such a circular root locus exists in this case, as shown in the following. The angle condition is

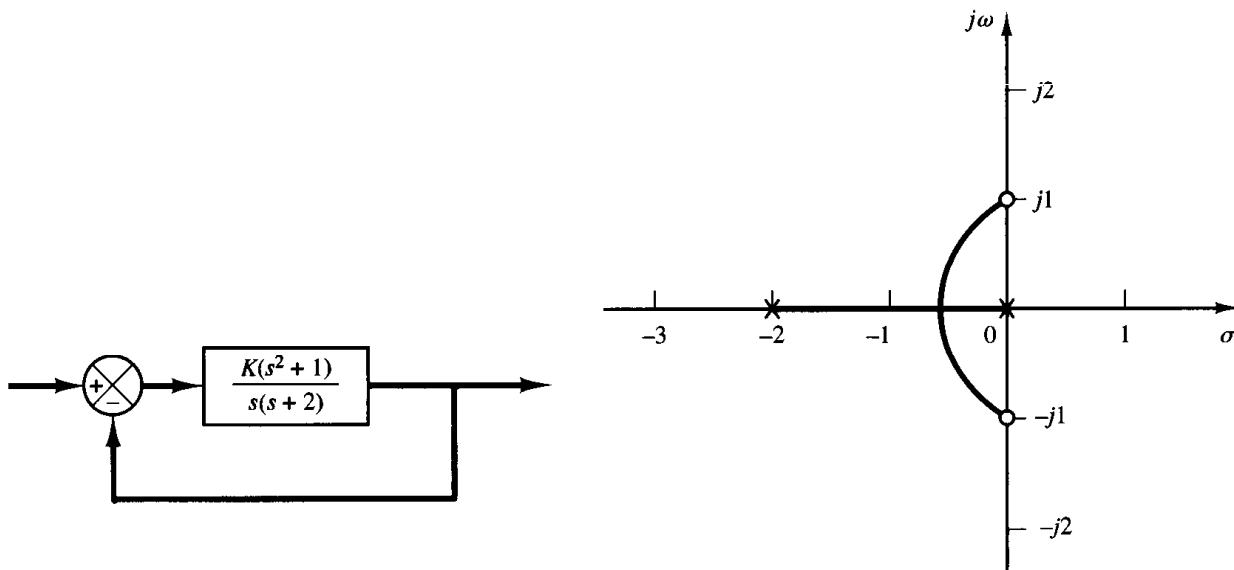


Figure 6-53
(a) Control system; (b) root-locus plot.

$$\angle \frac{K(s+j)(s-j)}{s(s+2)} = \pm 180^\circ(2k+1)$$

or

$$\angle s+j + \angle s-j - \angle s - \angle s+2 = \pm 180^\circ(2k+1)$$

By substituting $s = \sigma + j\omega$ into this last equation, we obtain

$$\angle \sigma + j\omega + j + \angle \sigma + j\omega - j = \angle \sigma + j\omega + \angle \sigma + 2 + j\omega \pm 180^\circ(2k+1)$$

or

$$\tan^{-1}\left(\frac{\omega+1}{\sigma}\right) + \tan^{-1}\left(\frac{\omega-1}{\sigma}\right) = \tan^{-1}\left(\frac{\omega}{\sigma}\right) + \tan^{-1}\left(\frac{\omega}{\sigma+2}\right) \pm 180^\circ(2k+1)$$

Taking tangents of both sides of this equation and noting that

$$\tan\left[\tan^{-1}\left(\frac{\omega}{\sigma+2}\right) \pm 180^\circ\right] = \frac{\omega}{\sigma+2}$$

we obtain

$$\frac{\frac{\omega+1}{\sigma} + \frac{\omega-1}{\sigma}}{1 - \frac{\omega+1}{\sigma} \frac{\omega-1}{\sigma}} = \frac{\frac{\omega}{\sigma} + \frac{\omega}{\sigma+2}}{1 - \frac{\omega}{\sigma} \frac{\omega}{\sigma+2}}$$

or

$$\omega\left[\left(\sigma - \frac{1}{2}\right)^2 + \omega^2 - \frac{5}{4}\right] = 0$$

which is equivalent to

$$\omega = 0 \quad \text{or} \quad \left(\sigma - \frac{1}{2}\right)^2 + \omega^2 = \frac{5}{4}$$

These two equations are equations for the root loci. The first equation corresponds to the root locus on the real axis. (The segment between $s = 0$ and $s = -2$ corresponds to the root locus for $0 \leq K < \infty$. The remaining parts of the real axis correspond to the root locus for $K < 0$.) The second equation is an equation for a circle. Thus, there exists a circular root locus with center at $\sigma = \frac{1}{2}$, $\omega = 0$ and the radius equal to $\sqrt{5/2}$. The root loci are sketched in Figure 6-53(b). [That part of the circular locus to the left of the imaginary zeros corresponds to $K > 0$. The portion of the circular locus not shown in Figure 6-53(b) corresponds to $K < 0$.]

- A-6-12.** Consider the system shown in Figure 6-54. Determine the value of α such that the damping ratio ζ of the dominant closed-loop poles is 0.5.

Solution. In this system the characteristic equation is

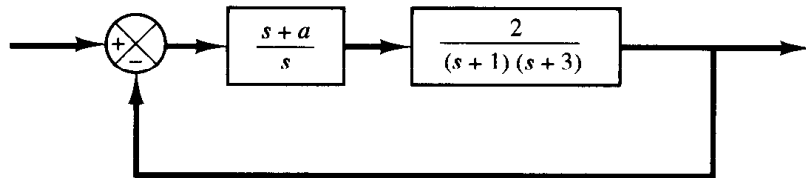


Figure 6-54
Control system.

$$1 + \frac{2(s + a)}{s(s + 1)(s + 3)} = 0$$

Notice that the variable a is not a multiplying factor. Hence, we need to rewrite the characteristic equation

$$s(s + 1)(s + 3) + 2s + 2a = 0$$

as follows:

$$1 + \frac{2a}{s^3 + 4s^2 + 5s} = 0$$

Define

$$2a = K$$

Then we get the characteristic equation in the form

$$1 + \frac{K}{s(s^2 + 4s + 5)} = 0 \quad (6-32)$$

In Problem A-6-4 we constructed the root-locus diagram for the system defined by Equation (6-32). Hence, the solution to this problem is available in Problem A-6-4. Referring to Figure 6-45(b), the closed-loop poles having the damping ratio $\zeta = 0.5$ can be located at $s = -0.63 \pm j1.09$. The value of K at point $s = -0.63 + j1.09$ may be found as 4.32. Hence, the value of a in this problem is obtained as follows:

$$a = \frac{K}{2} = 2.16$$

- A-6-13.** Consider the system shown in Figure 6-55(a). Determine the value of a such that the damping ratio ζ of the dominant closed poles is 0.5.

Solution. The characteristic equation is

$$1 + \frac{10(s + a)}{s(s + 1)(s + 8)} = 0$$

The variable a is not a multiplying factor. Hence, we need to modify the characteristic equation. Since the characteristic equation can be written as

$$s^3 + 9s^2 + 18s + 10a = 0$$

we rewrite this equation such that a appears as a multiplying factor as follows:

$$1 + \frac{10a}{s(s^2 + 9s + 18)} = 0$$

Define

$$10a = K$$

Then the characteristic equation becomes

$$1 + \frac{K}{s(s^2 + 9s + 18)} = 0$$

Notice that the characteristic equation is in a suitable form for the construction of the root loci.

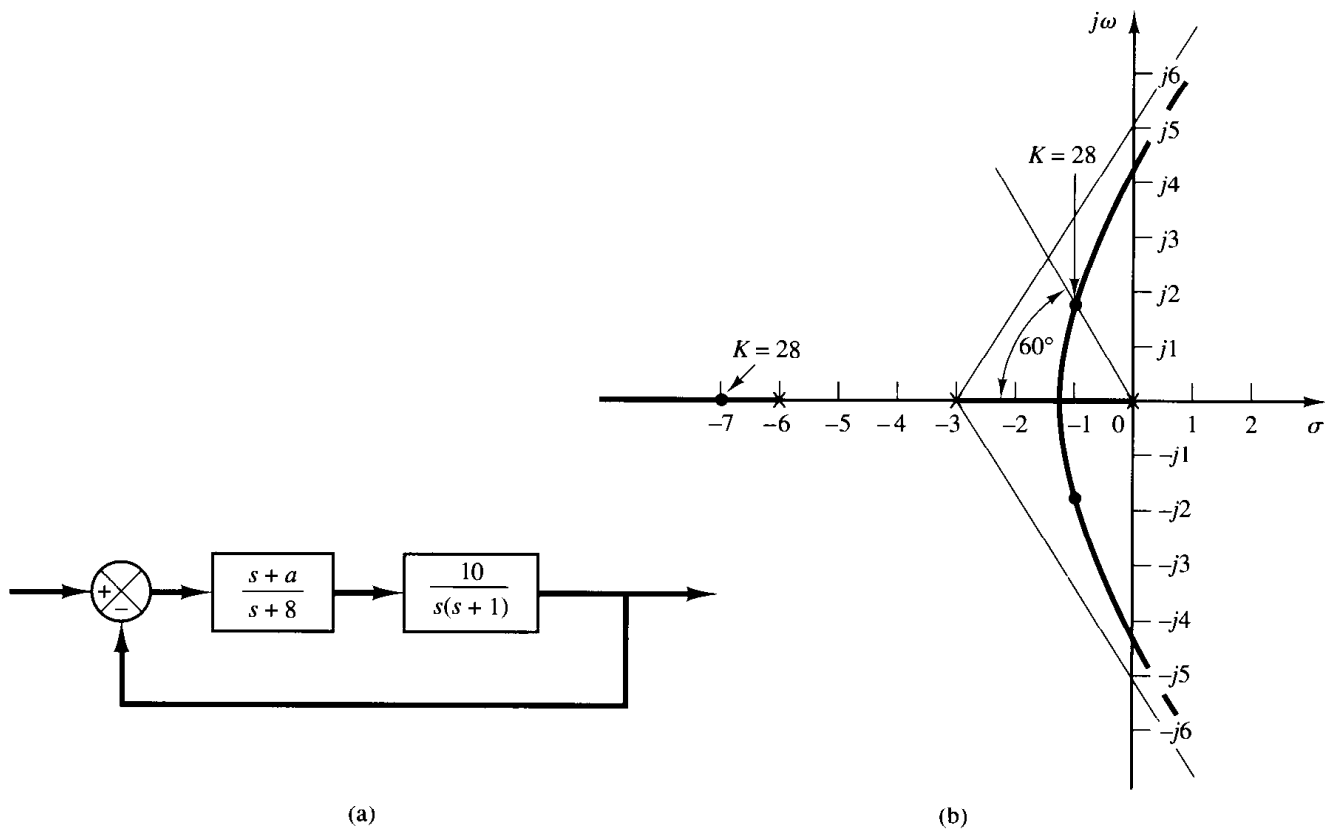


Figure 6-55

(a) Control system; (b) root-locus plot, where $K = 10a$.

This system involves three poles and no zero. The three poles are at $s = 0$, $s = -3$, and $s = -6$. A root-locus branch exists on the real axis between points $s = 0$ and $s = -3$. Also, another branch exists between points $s = -6$ and $s = -\infty$.

The asymptotes for the root loci are found as follows:

$$\text{Angles of asymptotes} = \frac{\pm 180^\circ(2k+1)}{3} = 60^\circ, -60^\circ, 180^\circ$$

The intersection of the asymptotes and the real axis is obtained from

$$\sigma_a = -\frac{0+3+6}{3} = -3$$

The breakaway and break-in points can be determined from $dK/ds = 0$, where

$$K = -(s^3 + 9s^2 + 18s)$$

Now we set

$$\frac{dK}{ds} = -(3s^2 + 18s + 18) = 0$$

which yields

$$s^2 + 6s + 6 = 0$$

or

$$s = -1.268, \quad s = -4.732$$

Point $s = -1.268$ is on a root-locus branch. Hence, point $s = -1.268$ is an actual breakaway point. But point $s = -4.732$ is not on the root locus and therefore is neither a breakaway nor break-in point.

Next we shall find points where root-locus branches cross the imaginary axis. We substitute $s = j\omega$ in the characteristic equation, which is

$$s^3 + 9s^2 + 18s + K = 0$$

as follows:

$$(j\omega)^3 + 9(j\omega)^2 + 18(j\omega) + K = 0$$

or

$$(K - 9\omega^2) + j\omega(18 - \omega^2) = 0$$

from which we get

$$\omega = \pm 3\sqrt{2}, \quad K = 9\omega^2 = 162 \quad \text{or} \quad \omega = 0, \quad K = 0$$

The crossing points are at $\omega = \pm 3\sqrt{2}$ and the corresponding value of gain K is 162. Also, a root-locus branch touches the imaginary axis at $\omega = 0$. Figure 6-55(b) shows a sketch of the root loci for the system.

Since the damping ratio of the dominant closed-loop poles is specified as 0.5, the desired closed-loop pole in the upper-half s plane is located at the intersection of the root-locus branch in the upper-half s plane and a straight line having an angle of 60° with the negative real axis. The desired dominant closed-loop poles are located at

$$s = -1 + j1.732, \quad s = -1 - j1.732$$

At these points, the value of gain K is 28. Hence,

$$a = \frac{K}{10} = 2.8$$

Since the system involves two or more poles than zeros (in fact, three poles and no zero), the third pole can be located on the negative real axis from the fact that the sum of the three closed-loop poles is -9 . Hence, the third pole is found to be at

$$s = -9 - (-1 + j1.732) - (-1 - j1.732)$$

or

$$s = -7$$

A-6-14. Consider the system shown in Figure 6-56(a). Sketch the root loci of the system as the velocity feedback gain k varies from zero to infinity. Determine the value of k such that the closed-loop poles have the damping ratio ζ of 0.7.

Solution. The open-loop transfer function is

$$\text{Open-loop transfer function} = \frac{10}{(s + 1 + 10k)s + 10}$$

Since k is not a multiplying factor, we modify the equation such that k appears as a multiplying factor. Since the characteristic equation is

$$s^2 + s + 10ks + 10 = 0$$

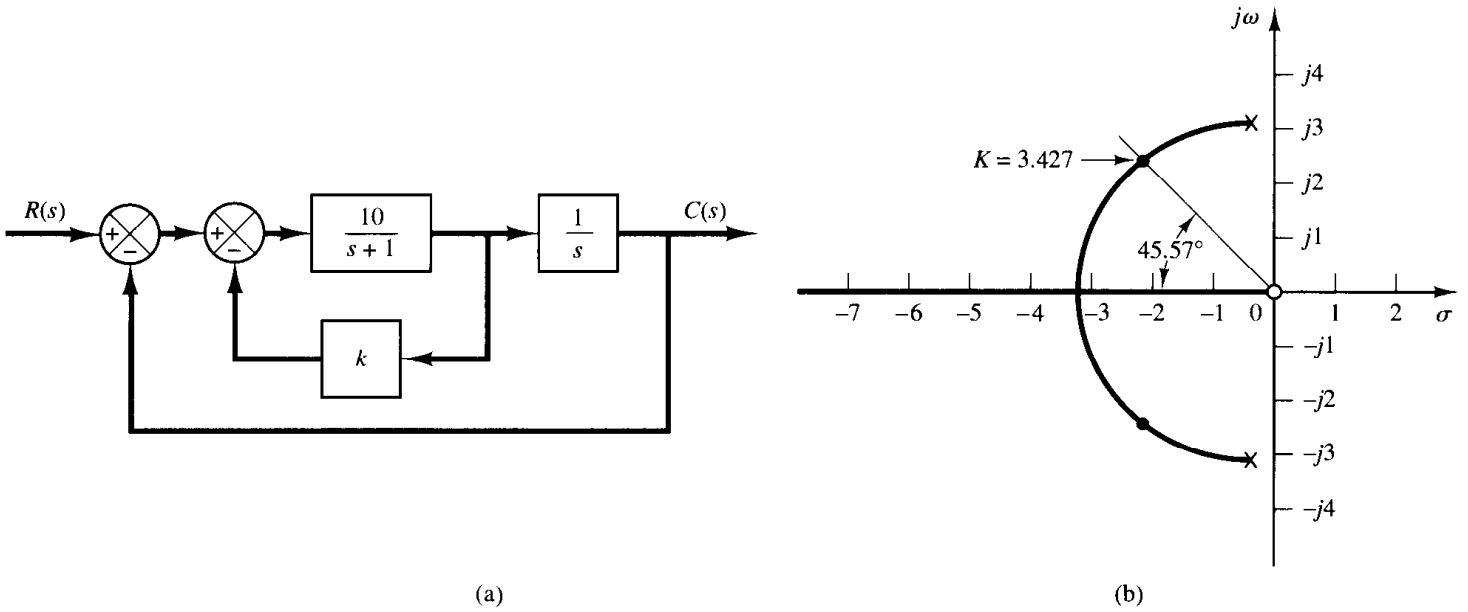


Figure 6-56

(a) Control system; (b) root-locus plot, where $K = 10k$.

we rewrite this equation as follows:

$$1 + \frac{10ks}{s^2 + s + 10} = 0 \quad (6-33)$$

Define

$$10k = K$$

Then Equation (6-33) becomes

$$1 + \frac{Ks}{s^2 + s + 10} = 0$$

Notice that the system has a zero at $s = 0$ and two poles at $s = -0.5 \pm j3.1225$. Since this system involves two poles and one zero, there is a possibility that a circular root locus exists. In fact, this system has a circular root locus, as will be shown. Since the angle condition is

$$\angle \frac{Ks}{s^2 + s + 10} = \pm 180^\circ(2k + 1)$$

we have

$$\angle s - \angle s + 0.5 + j3.1225 - \angle s + 0.5 - j3.1225 = \pm 180^\circ(2k + 1)$$

By substituting $s = \sigma + j\omega$ into this last equation and rearranging, we obtain

$$\angle \sigma + 0.5 + j(\omega + 3.1225) + \angle \sigma + 0.5 + j(\omega - 3.1225) = \angle \sigma + j\omega \pm 180^\circ(2k + 1)$$

which can be rewritten as

$$\tan^{-1} \left(\frac{\omega + 3.1225}{\sigma + 0.5} \right) + \tan^{-1} \left(\frac{\omega - 3.1225}{\sigma + 0.5} \right) = \tan^{-1} \left(\frac{\omega}{\sigma} \right) \pm 180^\circ(2k + 1)$$

Taking tangents of both sides of this last equation, we obtain

$$\frac{\frac{\omega + 3.1225}{\sigma + 0.5} + \frac{\omega - 3.1225}{\sigma + 0.5}}{1 - \left(\frac{\omega + 3.1225}{\sigma + 0.5}\right)\left(\frac{\omega - 3.1225}{\sigma + 0.5}\right)} = \frac{\omega}{\sigma}$$

which can be simplified to

$$\frac{2\omega(\sigma + 0.5)}{(\sigma + 0.5)^2 - (\omega^2 - 3.1225^2)} = \frac{\omega}{\sigma}$$

or

$$\omega(\sigma^2 - 10 + \omega^2) = 0$$

which yields

$$\omega = 0 \quad \text{or} \quad \sigma^2 + \omega^2 = 10$$

Notice that $\omega = 0$ corresponds to the real axis. The negative real axis (between $s = 0$ and $s = -\infty$) corresponds to $K \geq 0$, and the positive real axis corresponds to $K < 0$. The equation

$$\sigma^2 + \omega^2 = 10$$

is an equation of a circle with center at $\sigma = 0, \omega = 0$ with the radius equal to $\sqrt{10}$. A portion of this circle that lies to the left of the complex poles corresponds to the root locus for $K > 0$. The portion of the circle which lies to the right of the complex poles corresponds to the root locus for $K < 0$. Hence, this portion is not a root locus for the present system, where $K > 0$. Figure 6-56(b) shows a sketch of the root loci.

Since we require $\zeta = 0.7$ for the closed-loop poles, we find the intersection of the circular root locus and a line having an angle of 45.57° (note that $\cos 45.57^\circ = 0.7$) with the negative real axis. The intersection is at $s = -2.214 + j2.258$. The gain K corresponding to this point is 3.427. Hence, the desired value of the velocity feedback gain k is

$$k = \frac{K}{10} = 0.3427$$

A-6-15. Consider the control system shown in Figure 6-57. Plot root loci with MATLAB.

Solution. MATLAB Program 6-10 generates a root-locus plot as shown in Figure 6-58. The root loci must be symmetric about the real axis. However, Figure 6-58 shows otherwise.

MATLAB supplies its own set of gain values that are used to calculate a root-locus plot. It does so by an internal adaptive step-size routine. However, in certain systems, very small changes in the gain cause drastic changes in root locations within a certain range of gains. Thus, MATLAB takes too big a jump in its gain values when calculating the roots, and root locations change by a relatively large amount. When plotting, MATLAB connects these points and causes a strange looking graph at the location of sensitive gains. Such erroneous root-locus plots typically occur when the loci approach a double pole (or triple or higher pole), since the locus is very sensitive to small gain changes.

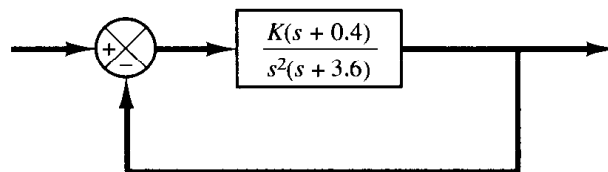


Figure 6-57
Control system.

MATLAB Program 6-10

```
% ----- Root-locus plot -----

num = [0 0 1 0.4];
den = [1 3.6 0 0];
rlocus(num,den);
v = [-5 1 -3 3]; axis(v)
grid
title('Root-Locus Plot of G(s) = K(s + 0.4)/[s^2(s + 3.6)]')
```

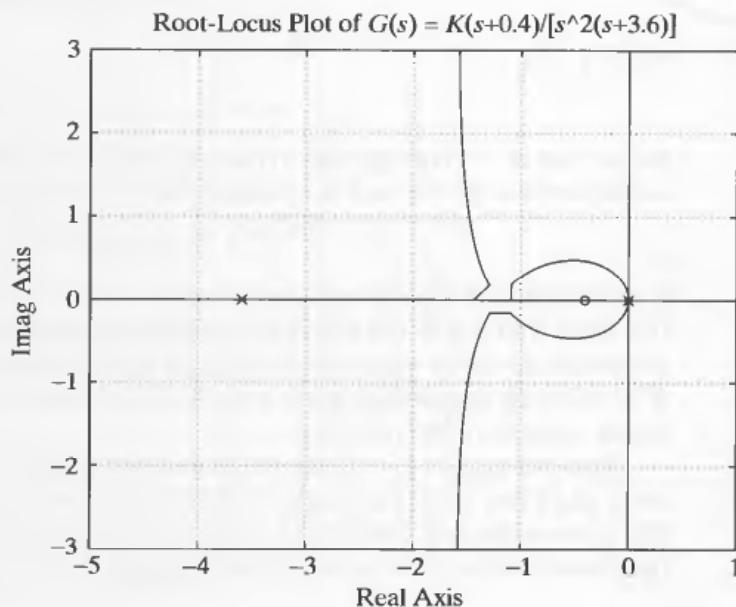


Figure 6-58
Root-locus plot.

In the problem considered here, the critical region of gain K is between 4.2 and 4.4. Thus we need to set the step size small enough in this region. We may divide the region for K as follows:

```
K1 = [0:0.2:4.2];
K2 = [4.2:0.002:4.4];
K3 = [4.4:0.2:10];
K4 = [10:5:200];
K = [K1 K2 K3 K4];
```

Entering MATLAB Program 6-11 into the computer, we obtain the plot as shown in Figure 6-59. If we change the plot command `plot(r,'o')` in MATLAB Program 6-11 to `plot(r,'-')`, we obtain Figure 6-60. Figures 6-59 and 6-60 respectively show satisfactory root-locus plots.

A-6-16. Consider the system whose open-loop transfer function $G(s)H(s)$ is given by

$$G(s)H(s) = \frac{K}{s(s+1)(s+2)}$$

Using MATLAB, plot root loci and their asymptotes.

MATLAB Program 6-11

% ----- Root-locus plot -----

```
num = [0 0 1 0.4];
den = [1 3.6 0 0];
K1 = [0:0.2:4.2];
K2 = [4.2:0.002:4.4];
K3 = [4.4:0.02:10];
K4 = [10:5:200];
K = [K1 K2 K3 K4];
r = rlocus(num,den,K);
plot(r,'o')
v = [-5 1 -5 5]; axis(v)
grid
title('Root-Locus Plot of G(s) = K(s + 0.4)/[s^2(s + 3.6)]')
xlabel('Real Axis')
ylabel('Imag Axis')
```

Solution. We shall plot the root loci and asymptotes on one diagram. Since the open-loop transfer function is given by

$$G(s)H(s) = \frac{K}{s(s+1)(s+2)}$$

$$= \frac{K}{s^3 + 3s^2 + 2s}$$

the equation for the asymptotes may be obtained as follows: Noting that

$$\lim_{s \rightarrow \infty} \frac{K}{s^3 + 3s^2 + 2s} \doteq \lim_{s \rightarrow \infty} \frac{K}{s^3 + 3s^2 + 3s + 1} = \frac{K}{(s+1)^3}$$

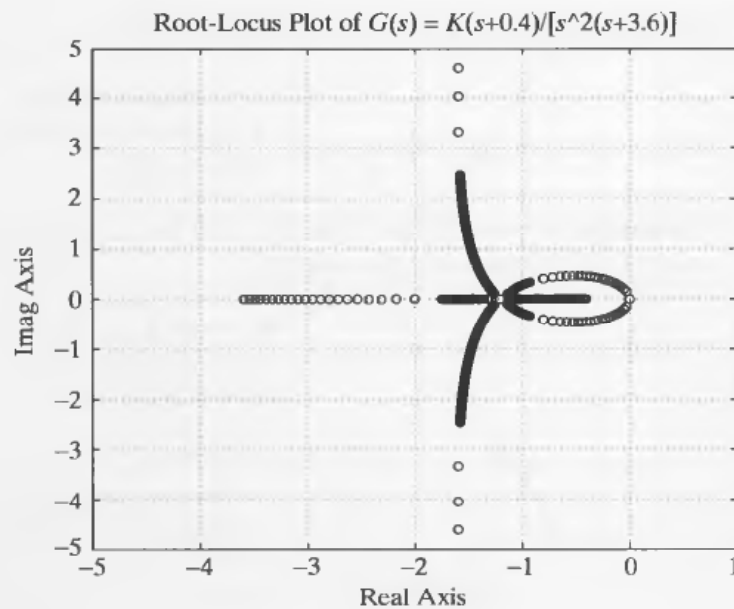


Figure 6-59
Root-locus plot.

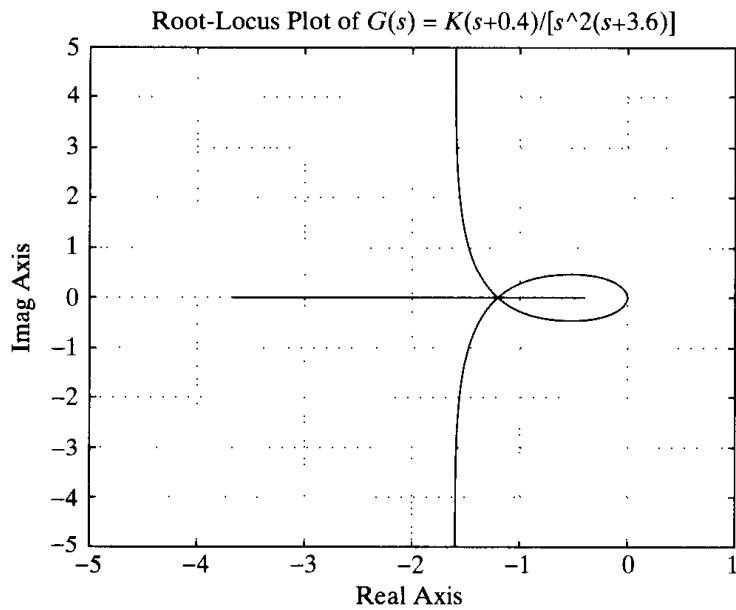


Figure 6-60
Root-locus plot.

the equation for the asymptotes may be given by

$$G_a(s)H_a(s) = \frac{K}{(s+1)^3}$$

Hence, for the system we have

$$\begin{aligned} \text{num} &= [0 \quad 0 \quad 0 \quad 1] \\ \text{den} &= [1 \quad 3 \quad 2 \quad 0] \end{aligned}$$

and for the asymptotes,

$$\begin{aligned} \text{numa} &= [0 \quad 0 \quad 0 \quad 1] \\ \text{dena} &= [1 \quad 3 \quad 3 \quad 1] \end{aligned}$$

In using the following root-locus and plot commands

```
r = rlocus(num,den)
a = rlocus(numa,dena)
plot([r a])
```

the number of rows of r and that of a must be the same. To ensure this, we include the gain constant K in the commands. For example,

```
K1 = 0:0.1:0.3;
K2 = 0.3:0.005:0.5;
K3 = 0.5:0.5:10;
K4 = 10:5:100;
K = [K1 K2 K3 K4]
r = rlocus(num,den,K)
a = rlocus(numa,dena,K)
y = [r a]
plot(y,'b')
```

Including gain K in rlocus command ensures that the r matrix and a matrix have the same number of rows. MATLAB Program 6–12 will generate a plot of root loci and their asymptotes. See Figure 6–61.

MATLAB Program 6–12

```
% ----- Root-Locus Plots -----

num = [0  0  0  1];
den = [1  3  2  0];
numa = [0  0  0  1];
dena = [1  3  3  1];
K1 = 0:0.1:0.3;
K2 = 0.3:0.005:0.5;
K3 = 0.5:0.5:10;
K4 = 10:5:100;
K = [K1  K2  K3  K4];
r = rlocus(num,den,K);
a = rlocus(numa,dena,K);
y = [r  a];
plot(y,'-')
v = [-4  4  -4  4]; axis(v)
grid
title('Root-Locus Plot of  $G(s) = K/[s(s+1)(s+2)]$  and Asymptotes')
xlabel('Real Axis')
ylabel('Imag Axis')

% ***** Manually draw open-loop poles in the hard copy *****
```

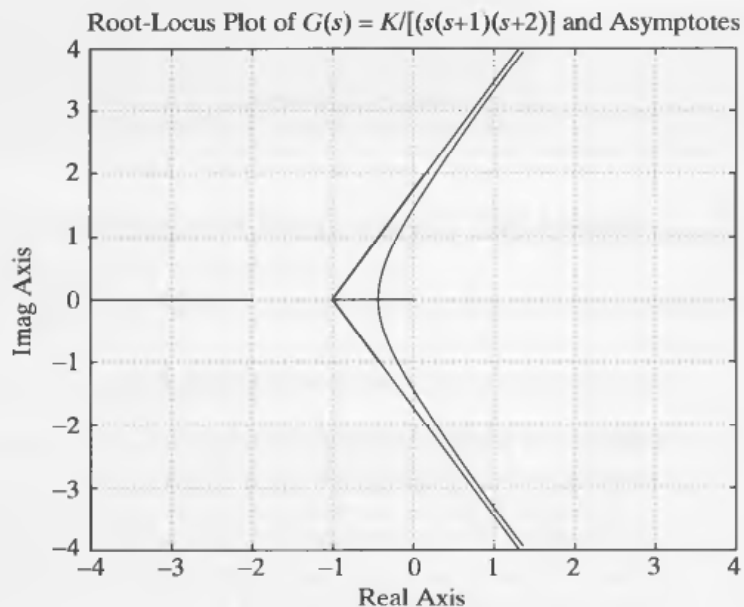


Figure 6–61
Root-locus plot.

Drawing two or more plots in one diagram can be accomplished by using the hold command. MATLAB Program 6–13 uses the hold command. The resulting root-locus plot is shown in Figure 6–62.

MATLAB Program 6–13

```
% ----- Root-Locus Plots -----

num = [0 0 0 1];
den = [1 3 2 0];
numa = [0 0 0 1];
dena = [1 3 3 1];
K1 = 0:0.1:0.3;
K2 = 0.3:0.005:0.5;
K3 = 0.5:0.5:10;
K4 = 10:5:100;
K = [K1 K2 K3 K4];
r = rlocus(num,den,K);
a = rlocus(numa,dena,K);
plot(r,'o')
hold
Current plot held
plot(a,'-')
v = [-4 4 -4 4]; axis(v)
grid
title('Root-Locus Plot of G(s) = K/[s(s + 1)(s + 2)] and Asymptotes')
xlabel('Real Axis')
ylabel('Imag Axis')

% ***** Manually draw open-loop poles in the hard copy *****

% ***** Remove hold on graphics *****

hold
Current plot released
```

A-6-17. Consider a unity-feedback system with the following feedforward transfer function $G(s)$:

$$G(s) = \frac{K(s + 2)^2}{(s^2 + 4)(s + 5)^2}$$

Plot root loci for the system with MATLAB.

Solution. A MATLAB program to plot the root loci is given as MATLAB Program 6–14. The resulting root-locus plot is shown in Figure 6–63.

Notice that this is a special case where no root locus exists on the real axis. This means that for any value of $K > 0$ the closed-loop poles of the system are two sets of complex-conjugate poles. (No real closed-loop poles exist.) Since no closed-loop poles exist in the right-half s plane, the system is stable for all values of $K > 0$.